When Does Safety Activate? Temporal Analysis of LLM Alignment via Logit-Margin Score

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Abstract

Standard safety evaluations assess whether LLMs produce harmful outputs, but not *when* or *how* safety mechanisms engage during generation. We introduce the **Logit-Margin Score** (S_t), a lightweight observability metric that tracks compliance–refusal balance throughout decoding using only standard logits. Applied to 660 harmful prompts across four models and three attack paradigms, generation-time summaries predict jailbreak outcomes (best ROC-AUC = 0.786 [0.750, 0.819]) while the activation latency t_{cross} quantifies *when* refusal dominates (AUC = 0.776). Temporal trajectory analysis reveals two distinct attack mechanisms invisible to output-level metrics: **Type A** (pre-generation context bias, where refusal signals still emerge but the attack operates upstream) and **Type B** (decoding-time interference, where refusal activation is suppressed during generation), classified via permutation tests with Benjamini–Hochberg FDR correction. Models exhibit qualitatively different refusal dynamics: Llama shows Type B under gradient-based attacks but Type A under multi-turn manipulation; Qwen under gradient-based attacks falls into an *Undetected* category where no measurable refusal response emerges. A token-efficient early stopping policy—halting generation if $S_0 < \tau$ (pre-generation gate) and no refusal signal emerges within k decoding steps—reduces benign false-halt rate from 41.1% to 13.8% while retaining 17.3 pp of attack reduction and improving preservation rate (83.2% $\rightarrow$ 91.0%), and its differential precision validates the taxonomy. Type A conditions show unchanged ΔASR (the policy correctly abstains because refusal signals emerge); Undetected conditions show the largest reduction, confirming genuine absence of refusal response.

1 Introduction

Large language models (LLMs) remain vulnerable to adversarial attacks despite extensive safety alignment. A growing body of work documents diverse attack strategies—including prompt engineering, gradient-based suffix optimization, and multi-turn context manipulation—that successfully elicit harmful outputs from aligned models [17, 24]. The standard metric, *Attack Success Rate* (ASR), answers only *what* happened, not *why*: two jailbreaks may both succeed under ASR yet exploit fundamentally different weaknesses—one biasing context *before* generation, another interfering with safety *during* decoding.

This limitation reflects a broader gap: existing evaluations collapse a temporally extended decoding process into a single binary label, obscuring *when* safety mechanisms activate and *how* they fail [13, 4]. Recent work has begun examining safety as a *process*—via representation directions [1, 23], decoding-time modification [18, 7], and reasoning-chain degradation [21]—but token-level temporal observability of refusal activation remains unexplored.

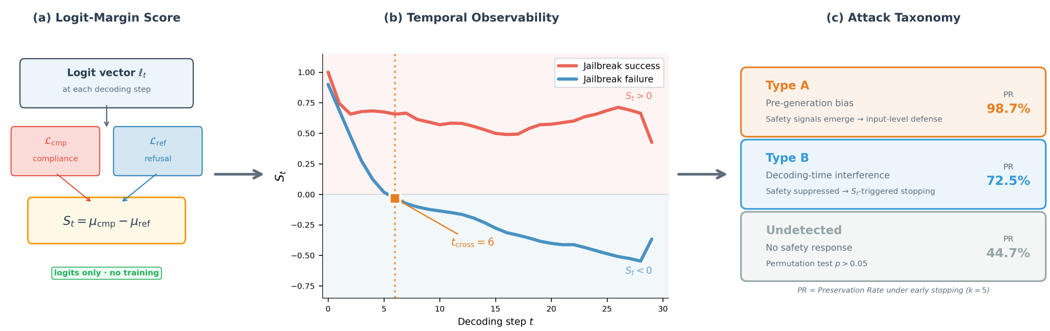

Figure 1: **Method overview.** (a) At each decoding step, logits are probed against compliance and refusal lexicons via top-k aggregation to yield S_t—requiring only standard logits, no gradients or hidden states. (b) S_t trajectories diverge between successful (red, stays positive) and failed (blue, crosses zero at t_{cross}) jailbreaks, exposing refusal dynamics invisible to output-level metrics. (c) Temporal signatures define three attack categories; preservation rate (PR) under k=5 early stopping validates the taxonomy.

We address this gap with the **Logit-Margin Score** (S_t), a temporal observability framework that tracks the compliance–refusal balance during autoregressive generation. Defined as the difference between aggregated compliance-token logits and refusal-token logits, S_t yields a continuous trajectory $S_0, S_1, \ldots, S_T$ that exposes the temporal structure of refusal activation and collapse (Figure 1). The framework requires only white-box logit access—no gradients, hidden states, or architectural modifications.

Using this trajectory-level view, we identify two qualitatively distinct attack signatures invisible to output-level metrics. *Type A* attacks operate primarily through pre-generation context bias: refusal signals still emerge during decoding (the model "attempts" refusal in logit space), but the biased context produces harmful output regardless. *Type B* attacks suppress refusal activation during decoding itself. This distinction maps onto different intervention strategies: input-level filtering for Type A and generation-time intervention for Type B. We operationalize this through a token-efficient early stopping policy: gating on $S_0 < \tau$ (pre-generation) and $t_{\text{cross}} > k$ (generation-time), it reduces the benign false-halt rate from 41.1% to 13.8% while retaining 68% of attack reduction, and its differential precision mirrors the taxonomy (Section 4.4).

Our contributions are fourfold:

1. A **temporal observability framework** based on S_t, providing token-level visibility into when and how refusal mechanisms engage during decoding. Notably, 91.6% of successful jailbreaks maintain compliance-dominant logits ($\bar{S} > 0$) throughout generation, confirming that temporal *dynamics*—not summary statistics alone—carry discriminative signal.
2. An **attack-mechanism taxonomy** (Type A, Type B, Undetected) via permutation tests with FDR correction, revealing distinct temporal signatures invisible to output-level evaluation.
3. **Construct validity**: generation-time summaries predict jailbreak outcomes (ROC-AUC = 0.786), and per-step correlation with hidden-state refusal directions ($|r|$ up to 0.62) confirms that S_t captures internal alignment representations.
4. **Token-efficient early stopping**: a two-stage policy combining $S_0 < \tau$ (pre-generation gate) with $t_{\text{cross}} > k$ (generation-time rule) reduces benign false-halt from 41.1% to 13.8% while improving preservation rate to 91.0%, simultaneously validating the taxonomy via differential precision across Type A/B/Undetected.

2 Related Work

Jailbreak attacks and defenses. Adversarial attacks on aligned LLMs span prompt-level manipulations [17, 16], optimization-based suffix attacks [24], and multi-turn erosion [10, 15]. Defenses include decoding-time interventions—SafeDecoding [18] modifies token probabilities, CARE [7] implements detect–rollback—and training-time robustification [22]. Our S_t trajectory directly integrates with such defenses: CARE's detection stage can use $S_t < \tau$ as an interpretable trigger, as we demonstrate in Section 4.4. Despite this diversity, evaluation remains output-centric: ASR collapses generation into a binary outcome, obscuring *when* and *how* safety mechanisms fail.

Refusal representations and temporal dynamics. Activation-level work identifies geometric refusal directions [1, 23], while logit-level analyses show compliance–refusal differences predict safety outcomes [3, 11]. These works typically analyze safety at a single layer or time point; our contribution extends to the *temporal* dimension. Refusal Falls Off a Cliff [21] shows that refusal behavior can collapse late in generation; our early-k analysis provides complementary logit-level evidence, revealing that safety-critical information concentrates in early tokens for some conditions while others exhibit progressive or absent activation. Unlike logit-gap steering approaches [11] that *modify* the logit distribution to redirect generation, our framework only *reads* S_t and makes binary halt/continue decisions—the model's decoding process remains unmodified when generation proceeds.

Inference-time safety observability. Recent approaches deploy learned reward monitors or satisficing controllers to enforce safety during generation [7]. These achieve strong defense performance but require per-model reward training and operate as black-box triggers. S_t provides an interpretable, training-free diagnostic signal that can serve as a lightweight trigger within such pipelines. The Type A/B taxonomy further enables *targeted* deployment rather than uniform application of a single defense strategy. Outcome labeling relies on LLM-as-judge; recent work [20] highlights judge variability, motivating our multi-judge validation (Appendix P).

3 Method

We present an observability framework for analyzing alignment signals in large language models by tracking the temporal evolution of compliance- and refusal-associated logits during generation.

3.1 Problem Setup

Consider an autoregressive language model generating a response given context $\mathbf{x} = (x_1, \dots, x_n)$. At each decoding step t, the model produces a logit vector $\boldsymbol{\ell}_t \in \mathbb{R}^{|\mathcal{V}|}$. We assume access to token-level logits during generation, without requiring gradients, hidden states, or architectural modifications (a white-box or developer-access setting; limitations discussed in Section 5). We operate on raw logits rather than probabilities because the softmax operation compresses logits into a probability simplex, obscuring relative confidence among non-selected tokens [11]. Raw logits preserve information about competing alternatives at each decoding step, enabling direct comparison between safety-relevant token groups.

3.2 Compliance and Refusal Lexicons

We define two disjoint token sets targeting well-documented behavioral signatures of instruction-tuned LLMs. The **refusal lexicon** $\mathcal{L}_{\text{ref}}$ targets apology/refusal markers (`"Sorry"`), policy references (`"policy"`), and harm descriptors (`"illegal", "harmful"`)—tokens that mechanistic analysis traces to a single refusal direction across 13 model families [1]. The **compliance lexicon** $\mathcal{L}_{\text{cmp}}$ targets affirmative uptake (`"Sure", "Here's"`), procedural onset (`"Step 1"`), and instructional framing (`"Tutorial"`)—patterns that jailbreak attacks explicitly optimize for [24, 14]. The full lexicon (26 refusal + 21 compliance entries) is in Appendix A; results are stable across lexicon variants (AUC within ± 0.02) including a minimal 5-token set (Appendix L).

3.3 Logit-Margin Score

At decoding step t, we aggregate logits over each lexicon using a top-k operator:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \quad \mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \tag{1}$$

where top-$k(\mathcal{L})$ denotes the k tokens in $\mathcal{L}$ with the highest logits at step t. The **Logit-Margin Score** is defined as:

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \tag{2}$$

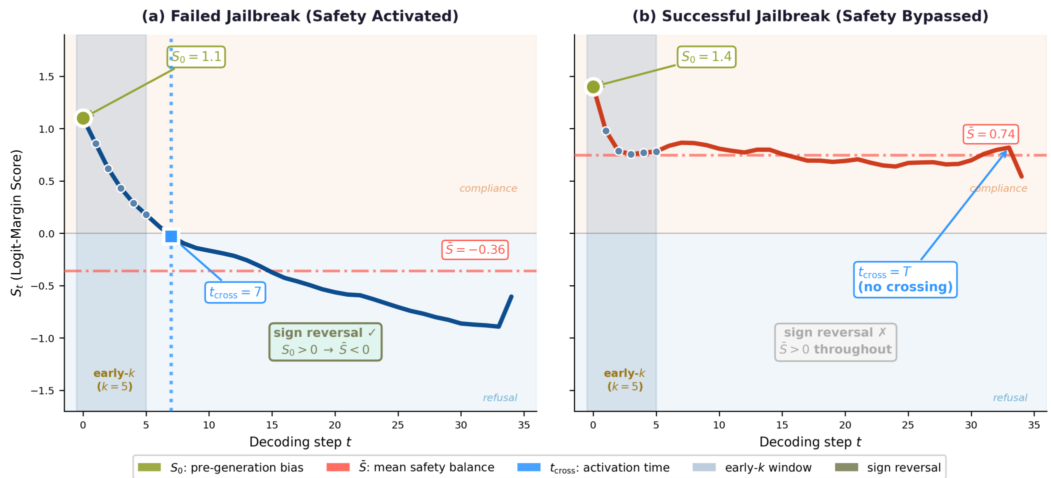

Figure 2: **Temporal metrics on S_t trajectories.** (a) **Failed jailbreak**: $S_0 > 0$, trajectory descends through the $\bar{S}_{1:k}$ window, crosses zero at $t_{\text{cross}} = 7$ (activation latency), and settles at $\bar{S} < 0$—producing a sign reversal. (b) **Successful jailbreak**: trajectory stays compliance-dominant ($\bar{S} > 0$), no zero-crossing occurs ($t_{\text{cross}} = T$), no sign reversal.

Table 1: Temporal metrics derived from $\{S_t\}_{t=0}^{T}$.

Metric	**Notation**	**Interpretation**
Pre-generation bias	S_0	Logit balance before decoding begins
Generation-time mean	$\bar{S}$	Average compliance–refusal balance
Sign reversal	$S_0 > 0 \wedge \bar{S} < 0$	Refusal activation indicator
Early-window mean	$\bar{S}_{1:k}$	Mean score over first k tokens
Activation latency	t_{cross}	First step at which $S_t < 0$

Positive S_t indicates a compliance-dominant state; negative S_t indicates a refusal-dominant state. The differential formulation cancels shared logit-magnitude variation ($r \approx 0.98$ between μ_{cmp} and μ_{ref}), isolating the alignment-specific signal (Appendix M). Per-step correlation with hidden-state refusal directions confirms alignment with internal representations ($|r|$ up to 0.62; Appendix R). S_t is independent of outcome labeling: many successful jailbreaks exhibit positive S_t despite producing harmful content (Appendix H). Figure 1 illustrates the computation pipeline and attack mechanism decomposition.

3.4 Temporal Metrics

From the sequence $\{S_t\}_{t=0}^{T}$, we derive scalar summaries (Table 1). S_0 is computed at the *final token position of the prompt*, before any response tokens are generated. We define **compliance-to-refusal transition** ("sign reversal") as $S_0 > 0 \wedge \bar{S} < 0$: the model begins in a compliance-dominant state but shifts to refusal dominance during generation, indicating refusal activation. Sign reversal is computed only for harmful samples; benign samples show format-dependent rather than safety-driven patterns (Appendix B.2). Each condition uses **format-matched baselines** to isolate harmful content effects from format effects.

Activation latency (t_{cross}). The activation latency is the first generation step at which refusal dominates:

$$t_{\text{cross}} = \min\{t \geq 1 : S_t < 0\}, \tag{3}$$

where $t \geq 1$ excludes S_0 (pre-generation). If no crossing occurs, $t_{\text{cross}} = T$ (Appendix F.5 verifies robustness to oscillation).

Relationship between $\bar{S}_{1:k}$ and t_{cross}. These two metrics capture complementary views of the same early-generation window. $\bar{S}_{1:k}$ is a continuous summary: how compliance-dominant the first k

tokens are on average. $t_{\text{cross}} > k$ is its binary counterpart: whether *any* safety signal appeared within those k steps. Their similar predictive power (AUC $= 0.786$ vs. 0.776) reflects this overlap. We retain both: $\bar{S}_{1:k}$ serves as a graded predictor, while $t_{\text{cross}} > k$ provides the threshold-free decision rule used for early stopping (Section 4.4).

3.5 Attack Taxonomy (Type A/B)

For comparative analysis across conditions, we define the **Relative Position (RP)** metric:

$$\text{RP}(M) = \frac{(M_{\text{success}} - M_{\text{harmful}})}{(M_{\text{benign}} - M_{\text{harmful}})}\,, \tag{4}$$

where M_{success}, M_{harmful}, and M_{benign} are condition means over successful attacks, all harmful samples, and format-matched benign queries, respectively. RP $= 0$ indicates success resembles the harmful distribution; RP $= 1$ indicates benign-like behavior. Crucially, RP does not require failure samples, enabling analysis under high ASR.

Type A/B classification. We apply RP to S_0 (RP$_A$) and $\bar{S}$ (RP$_B$), classifying conditions as:

$$\text{Type A} : |\text{RP}_A| > |\text{RP}_B|, \quad \text{Type B} : |\text{RP}_A| < |\text{RP}_B|. \tag{5}$$

Type A: the attack's primary effect is on the pre-generation logit state; refusal signals typically still emerge during decoding, but the context-level manipulation produces harmful output regardless—requiring input-level defense. **Type B**: the attack suppresses or delays refusal activation during decoding, making it amenable to S_t-triggered intervention. The magnitude $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$ indicates overall signal strength.

Undetected category. Conditions where r is not statistically significant are classified as *Undetected*. We test via label permutation (B = 10,000) with Benjamini–Hochberg FDR correction ($q = 0.05$, $m = 11$). Three conditions are Undetected: Qwen+GCG ($r = 0.01$, $p = 0.98$), Mistral+DeepInception ($p = 0.52$), and Mistral+MCM ($p = 0.061$; borderline). All eight significant conditions survive FDR correction (Table 4).

4 Experiments

We evaluate the Logit-Margin Score across three jailbreak paradigms and four model families, addressing: (i) predictive validity, (ii) temporal characterization, (iii) Type A/B analysis, and (iv) S_0-gated early stopping.

4.1 Setup

Models and attacks. We evaluate on Llama-3.1-8B-Instruct [6], Mistral-7B-Instruct-v0.3 [9], Qwen2.5-7B-Instruct [19], and Gemma-2-9B-Instruct [5] using greedy decoding. Attack paradigms: (1) **Multi-turn context manipulation (MCM)**: four-turn protocol with incrementally manipulated context; (2) **Gradient-based suffix (GCG)** [24]: adversarial suffix optimization; (3) **DeepInception** [12]: nested fictional scenario prompting (applied to Llama, Mistral, Qwen).

Dataset. We evaluate 660 harmful prompts (60 per model–attack condition) from JailbreakBench [2], with 660 attack-matched benign queries (11 conditions total; Gemma excludes DeepInception). Effect sizes are large (Cohen's $d > 1.0$, $p < 10^{-38}$). Outcome labels use **Llama-Guard-3-8B** [8] as primary LLM-as-judge, validated with HarmBench classifier (90% agreement) and GPT-4o (Appendix P). Table 2 summarizes composition and ASR.

4.2 Predictive Validation

On the combined harmful dataset ($n = 660$), $\bar{S}_{1:5}$ achieves ROC-AUC $= 0.786$ $[0.750, 0.819]$, followed by t_{cross} (0.776) and $\bar{S}$ (0.772); pre-generation S_0 alone reaches 0.696 (Table 3).

Table 2: Dataset composition. ASR via LLM-as-judge. ‡Benign GCG uses single-turn format.

Model	Attack	n	ASR	S_0	$\bar{S}$
Llama	MCM	60	8.3%	1.45	-1.42
Llama	GCG	60	15.0%	0.62	-0.99
Llama	DeepInception	60	56.7%	4.43	-0.11
Mistral	MCM	60	5.0%	4.51	-0.31
Mistral	GCG	60	56.7%	3.14	$+0.16$
Mistral	DeepInception	60	76.7%	3.04	$+0.43$
Qwen	MCM	60	51.7%	11.97	-0.12
Qwen	GCG	60	81.7%	7.13	$+0.71$
Qwen	DeepInception	60	55.0%	7.86	$+1.18$
Gemma	MCM	60	20.0%	4.67	$+0.47$
Gemma	GCG	60	33.3%	4.46	$+0.68$
Benign (matched)		660	—	varies by model	

Table 3: ROC-AUC comparison (LLM-as-judge, $n = 660$). Single authoritative reference for all AUC values reported in this paper.

Metric	AUC	95% CI
$\bar{S}_{1:5}$	**0.786**	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
$\bar{S}$	0.772	[0.736, 0.806]
$\bar{S}_{1:3}$	0.732	[0.692, 0.769]
Keyword baseline	0.732	[0.699, 0.764]
S_0	0.696	[0.656, 0.734]

Temporal patterns. Safety-relevant signal concentrates in the first few decoding tokens: AUC increases from 0.696 ($\bar{S}_{1:1}$) to 0.786 ($\bar{S}_{1:5}$), already approaching full-trajectory $\bar{S}$ (0.772). t_{cross} directly answers "when does refusal activate?": successful attacks delay activation (8.4 ± 6.1 steps) versus failures (4.0 ± 3.8; $d = 0.89$). Failed jailbreaks exhibit sign reversal $5.1\times$ more often than successes (44.3% vs. 8.7%). This concentration of signal in early tokens motivates token-efficient early stopping (Section 4.4).

4.3 Attack Paradigm Comparison (Type A/B Analysis)

We classify conditions by comparing $|\text{RP}_A|$ versus $|\text{RP}_B|$ among conditions where r is statistically significant (permutation $p < 0.05$, with Benjamini–Hochberg FDR correction for 11 simultaneous tests at $q = 0.05$); non-significant conditions are classified as Undetected. All eight significant conditions survive FDR correction; the three Undetected conditions remain non-significant. Table 4 reports RP values with permutation p-values and bootstrap 95% CIs.

Key findings. Attack mechanisms are model–attack specific rather than universal. **Llama** shows Type B for GCG and DeepInception but Type A for MCM—the same model engages different refusal dynamics depending on attack paradigm. **Qwen** is Type A for MCM but definitively Undetected for GCG ($r = 0.01$, $p = 0.98$). **Gemma** is consistently Type B; **Mistral** shows mixed patterns including Undetected under DeepInception despite 76.7% ASR (Appendix B.3).

Compliance-dominant jailbreaks. 91.6% of successful jailbreaks maintain $\bar{S} > 0$ throughout generation, meaning compliance tokens dominate even as the model produces harmful output. This confirms that S_t's discriminative power lies in its *temporal dynamics*—when refusal signals emerge (t_{cross}) and whether a sign reversal occurs—not in summary-level polarity. It also explains why the Type A/B taxonomy is essential: output-level metrics cannot distinguish "safety activated but overwhelmed" (Type A) from "safety never engaged" (Undetected).

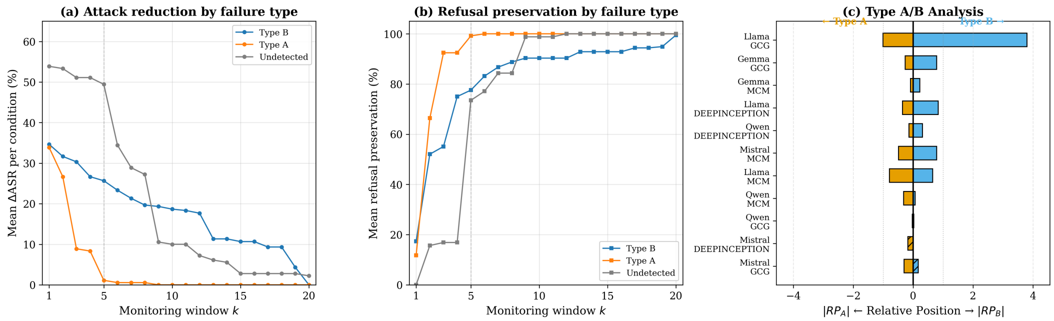

Figure 3: **Taxonomy validation.** (a) ΔASR by taxonomy category vs. observation window k: Type A drops to zero by k=5; Type B shows sustained reduction; Undetected declines steeply. (b) Preservation rate: Type A reaches $\approx$100% by k=3, confirming that safety activates and the policy correctly abstains. (c) Type A/B diverging bar: left $|\text{RP}_A|$, right $|\text{RP}_B|$; hatching = inverted. Dashed line: k=5.

Table 4: Type A/B analysis with permutation test and BH-FDR correction ($q = 0.05$, $m = 11$). p-values via label permutation ($B = 10{,}000$); 95% CIs via bootstrap.

Model + Attack	ASR	RP_A	RP_B	r	p_{perm}	95% CI(r)	Category
Llama + GCG	15.0%	0.66	2.89	2.96	<.001	[1.40, 4.71]	**Type B**
Llama + MCM	8.3%	0.88	0.69	1.12	<.001	[1.01, 1.25]	**Type A**
Llama + DeepInc.	56.7%	0.43	0.83	0.93	<.001	[0.60, 1.22]	**Type B**
Mistral + GCG	56.7%	0.30	−0.17	0.35	<.001	[0.17, 0.57]	**Type A**‡
Mistral + MCM	5.0%	0.49	0.79	0.93	.061	[0.44, 1.48]	*Undet.*§
Mistral + DeepInc.	76.7%	−0.19	0.00	0.19	.517	[0.07, 1.65]	*Undetected*
Qwen + MCM	51.7%	0.57	0.12	0.58	<.001	[0.14, 1.20]	**Type A**
Qwen + GCG	81.7%	0.00	−0.01	0.01	.981	[0.02, 0.20]	*Undetected*†
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	.029	[0.14, 0.77]	**Type B**
Gemma + GCG	33.3%	0.27	0.79	0.83	.003	[0.34, 1.41]	**Type B**
Gemma + MCM	20.0%	0.09	0.22	0.24	<.001	[0.14, 0.31]	**Type B**

†Strongest Undetected: $p = 0.98$, convergent with t_{cross} AUC $= 0.50$. ‡Negative RP: inverted pattern. §Borderline: large r but underpowered ($n_{\text{success}} = 3$).

4.4 Token-Efficient Early Stopping via t_{cross}

A direct practical implication of temporal observability is *token efficiency*: if the first k decoding steps reveal no refusal activation (S_t remains positive), the generation is unlikely to self-correct—halting early avoids producing the remaining $T - k$ tokens of potentially harmful output. We implement a threshold-free early stopping rule: halt generation if $t_{\text{cross}} > k$ (no zero-crossing within k steps), replacing the output with a standard refusal. The decision boundary is the natural sign change of S_t; the only hyperparameter is k. Unlike logit-gap steering [11] that modifies logits, this policy only *reads* S_t. We set $k = 5$ and evaluate via simulation: samples with $t_{\text{cross}} > 5$ are relabeled as halted. Beyond efficiency, this experiment serves a second purpose: the *differential precision* across taxonomy categories directly tests whether Type A/B/Undetected classifications capture genuine mechanistic differences.

Results. Table 5 and Figure 3 summarize the results. With only $k = 5$ tokens of observation, the policy reduces aggregate ASR from 39.7% to 14.2% while preserving 83.2% of successful refusals—demonstrating that discriminative signal is concentrated in the earliest decoding steps. The *differential precision* across taxonomy categories validates the classification: **Type A** conditions produce $t_{\text{cross}} \leq k$ because refusal signals *do* emerge during decoding—the policy correctly abstains (PR $= 98.7\%$), confirming that Type A requires input-level defense rather than generation-time intervention. **Type B** conditions show the primary efficiency gain (ΔASR $= 25.7$ pp, PR $= 72.5\%$), as the policy catches delayed refusal activation. **Undetected** conditions show the largest ΔASR (49.4 pp) but lowest PR (44.7%)—the policy halts indiscriminately because no discriminative signal exists.

Table 5: t_{cross}-gated early stopping ($k = 5$, Llama-Guard). PR: preservation rate. The taxonomy predicts intervention precision: Type A (refusal signals present) shows near-perfect PR; Undetected shows indiscriminate halting. $^{\diamond}\text{ASR}_{\text{orig}}$ values may differ from Table 2 because the simulation pipeline uses a label snapshot that includes minor judge-version updates; all within-table comparisons (ASR_{orig} vs. ASR_{int}) use identical labels.

Condition	Type	ASR_{orig}	ASR_{int}	ΔASR	PR
Gemma + GCG	**B**	33.3%	11.7%	21.7%	50.0%
Gemma + MCM	**B**	18.3%	18.3%	0.0%	95.9%
Llama + DeepInc.	**B**	55.0%	8.3%	46.7%	70.4%
Llama + GCG	**B**	8.3%	1.7%	6.7%	98.2%
Llama + MCM	**A**	5.0%	5.0%	0.0%	100.0%
Mistral + GCG	**A**	56.7%	53.3%	3.3%	96.2%
Mistral + DeepInc.	*U*	80.0%	0.0%	80.0%	0.0%
Mistral + MCM	*U*	5.0%	5.0%	0.0%	98.2%
Qwen + DeepInc.	**B**	58.3%	5.0%	53.3%	48.0%
Qwen + GCG	*U*	76.7%	8.3%	68.3%	35.7%
Qwen + MCM	**A**	40.0%	40.0%	0.0%	100.0%
Aggregated	–	39.7%	14.2%	**25.5%**	83.2%

Table 6: S_0-gated vs. original early stopping ($k = 5$, $\tau = 6.3$). Gating on $S_0 < \tau$ before applying $t_{\text{cross}} > k$ reduces benign false-halt from 41.1% to 13.8% while retaining 68% of ΔASR.

	Original ($t_{\text{cross}} > k$)		S_0**-Gated**	
	ΔASR	False-halt	ΔASR	False-halt
Aggregate	25.5%	41.1%	17.3%	13.8%
Gemma	10.8%	6.7%	7.5%	0.0%
Llama	17.8%	52.8%	14.4%	6.1%
Mistral	27.8%	45.6%	27.8%	44.4%
Qwen	40.6%	47.8%	16.1%	0.0%

This three-way pattern—*abstaining* on Type A, *catching* Type B, *indiscriminate* on Undetected—confirms that the taxonomy captures genuine differences in refusal dynamics.

S_0**-gated early stopping.** The unmodified policy yields a 41.1% benign false-halt rate because it cannot distinguish "harmful but no refusal" from "benign and no refusal needed." We address this by gating on S_0: since low S_0 indicates a refusal-leaning pre-generation state—more common for harmful queries—we apply early stopping only when $S_0 < \tau$:

$$\text{halt} \iff (S_0 < \tau) \ \wedge \ (t_{\text{cross}} > k). \tag{6}$$

This adds no computation (S_0 is already extracted) and mirrors the Type A/B structure: S_0 captures the pre-generation axis, t_{cross} the generation-time axis. Table 6 compares both policies with $\tau = 6.3$ (selected to maximize ΔASR subject to false-halt $\leq 15\%$). S_0-gating reduces benign false-halt from 41.1% to **13.8%** while retaining 17.3 of 25.5 pp aggregate ΔASR (68%) and *improving* preservation rate from 83.2% to 91.0%. The effect is model-dependent: Llama (52.8% $\rightarrow$ 6.1%) and Qwen (47.8% $\rightarrow$ 0.0%) improve dramatically because their benign S_0 is high; Mistral shows minimal change (45.6% $\rightarrow$ 44.4%) because its benign and harmful S_0 distributions overlap—a model-specific calibration difference that itself constitutes a diagnostic finding.

Multi-judge results are consistent (Appendix P.5).

4.5 Robustness

Results are robust across stochastic decoding (ICC = 0.92–1.0; Appendix G), baselines (fixed metrics capture 93% of learnable signal; Appendix K), and lexicon variants (AUC within ± 0.02; Appendix L). Hidden-state methods achieve 2–8 pp higher AUC but require activation access (Appendix E); data-driven lexicon induction confirms the handcrafted lexicon as a conservative lower bound (Appendix N).

5 Limitations and Discussion

Lexicon dependence. The Logit-Margin Score relies on manually constructed English lexicons. Results are stable across variants including a minimal 5-token set (AUC within ± 0.02; Appendix L), and probe-derived validation confirms alignment with each model's internal refusal direction (Appendix X). Style confounding is non-negligible (41.1% benign false-halt rate), but three convergent checks—token rank linkage, hidden-state correlation ($|r|$ up to 0.62), and common-mode rejection—confirm that $S_t < 0$ reflects genuine refusal activation rather than polite hedging (Appendix U, R). Multilingual extension is untested; data-driven lexicon induction offers a principled path (Appendix N).

Temporal stationarity. Lexicons are applied identically at every step despite position-dependent relevance. Diagnostic signal concentrates in early tokens ($|r|$ up to 0.62 at step 0, attenuating beyond step 10); lag-optimized analysis reaches $|r| = 0.42$–0.52 (Appendix R). Layer-sweep analysis reveals that S_t precedes hidden-state shifts at mid-network layers (12–16) but follows at early layers (8), converging to synchrony at layer 24—consistent with the early stopping policy targeting $k = 5$ tokens.

Early stopping scope. The S_0-gated policy reduces benign false-halt to 13.8% but remains a diagnostic demonstration, not a comprehensive defense. An adaptive adversary could manipulate S_0 or produce transient zero-crossings within k steps. The policy is by design ineffective for Type A and shows model-dependent gating efficacy (Mistral benign S_0 overlaps with harmful, yielding minimal FHR reduction). Prospective evaluation with generation halting remains future work; integration with CARE-style rollback [7] or SafeDecoding [18] could yield stronger defenses.

Outcome labeling and scope. Llama-Guard-3-8B labels are validated with HarmBench and GPT-4o; Type A/B classifications match in 8–9/10 conditions across judges (Appendix P). White-box logit access limits applicability to open-weight models; API-level access ($k = 20$) is insufficient (Appendix T). We evaluate three attack paradigms and four models; prompt injection, multilingual attacks, and closed-source models remain unexplored.

6 Conclusion

Refusal behavior in large language models is not a static gate but a temporal process that unfolds during decoding. The Logit-Margin Score makes this process observable at token-level granularity, without model modification or hidden-state access.

Across four models and three attack paradigms, S_t summaries predict jailbreak outcomes (AUC $= 0.786$) and the Type A/B decomposition—validated via permutation tests with FDR correction—reveals attack-specific temporal signatures invisible to output-level ASR. Notably, 91.6% of successful jailbreaks maintain compliance-dominant logits throughout generation, confirming that temporal *dynamics*—not summary statistics alone—carry discriminative signal. An S_0-gated early stopping policy demonstrates practical token efficiency: gating on $S_0 < \tau$ before applying $t_{\text{cross}} > k$ reduces benign false-halt from 41.1% to 13.8% while retaining 68% of attack reduction and improving preservation rate to 91.0%—and its differential precision mirrors the taxonomy, confirming that temporal diagnostics yield actionable predictions about where different defense strategies will succeed.

Outlook. Near-term extensions include: (i) integration with more sophisticated defense pipelines (CARE-style rollback, SafeDecoding); (ii) black-box application via provider-exposed log-probabilities; (iii) multilingual extension via data-driven lexicon induction (Appendix N); and (iv) temporally adaptive lexicons addressing position-dependent token relevance. More broadly, combining S_t with activation-steering methods could enable "sense-and-intervene" pipelines where lightweight temporal diagnostics trigger targeted, mechanistically-informed intervention.

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A Lexicon Specification

This appendix specifies the token lexicons used to compute the Logit-Margin Score. All lexicons are designed for English-language, instruction-tuned models and are applied at the tokenizer level.

Refusal Lexicon ($\mathcal{L}_{\text{ref}}$, 26 entries). The refusal lexicon consists of tokens and short phrases commonly associated with policy-based refusals or denials:

```
"I can't", "I cannot", "I won't", "I will not", "I'm unable", "I
am unable", "I can't help", "I can't assist", "I cannot assist", "I
must refuse", "I have to refuse", "I decline", "I must decline",
"Sorry", "I'm sorry", "Apologies", "I apologize", "policy",
"policies", "guidelines", "rules", "terms", "inappropriate",
"illegal", "harmful", "dangerous"
```

Compliance Lexicon ($\mathcal{L}_{\text{cmp}}$, 21 entries). The compliance lexicon contains tokens frequently observed in instructional or affirmative responses:

```
"Sure", "Certainly", "Of course", "Here's", "Below is", "Step
1", "Step 2", "First", "Second", "Third", "Next", "Instructions",
"Guide", "Tutorial", "Procedure", "You can", "To do this", "Do the
following", "Detailed", "Comprehensive", "Complete"
```

Implementation details. Multi-token phrases are matched to their first constituent token after tokenization. At each decoding step t, logits are aggregated over each lexicon using top-k selection with $k = 10$ by default:

$$\mu_{\text{cmp}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{cmp}})} \ell_t(v), \tag{7}$$

$$\mu_{\text{ref}}(t) = \frac{1}{k} \sum_{v \in \text{top-}k(\mathcal{L}_{\text{ref}})} \ell_t(v), \tag{8}$$

$$S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t). \tag{9}$$

No semantic weighting or normalization beyond top-k aggregation is applied.

B Additional Method Details

B.1 Why Logits?

Raw logits preserve information about competing alternatives at each decoding step that the softmax operation compresses into probabilities, obscuring relative confidence among non-selected tokens. This allows direct comparison between safety-relevant token groups [11] without requiring hidden-state access or architectural modifications.

B.2 Benign Sample Exclusion Rationale

Sign reversal is computed only for harmful samples because benign samples exhibit paradigm-dependent baseline behavior: MCM benign samples consistently show $\bar{S} > 0$ (0% sign reversal), while GCG benign samples frequently exhibit negative $\bar{S}$ (91.2% sign reversal). This discrepancy reflects *structural format differences*—single-turn GCG versus multi-turn MCM—rather than refusal activation. ANOVA confirms format explains ∼60% of variance in benign $\bar{S}$ ($F = 89.3$, $p < 10^{-15}$). To enable cross-paradigm comparison, we use **format-matched baselines**: each condition's benign baseline uses the same conversational structure, isolating harmful content effects from format effects.

B.3 RP Interpretation and Edge Cases

RP is an affine rescaling measuring where successful attacks lie along the harmful–benign continuum. Values outside $[0, 1]$ indicate attacks beyond the reference interval: RP > 1 means successful attacks "overshoot" benign (more compliance-like than typical benign queries), while RP < 0 means attacks lie beyond the harmful baseline—an inverted pattern suggesting the attack suppresses rather than elevates compliance signals. For reference, we also compute polar coordinates $\theta = \text{atan2}(\text{RP}_B, \text{RP}_A)$ and $r = \sqrt{\text{RP}_A^2 + \text{RP}_B^2}$. The diverging bar visualization (Figure 3c) distinguishes negative RP cases via hatching.

Undetected category. When $r < 0.1$, successful attacks are statistically indistinguishable from the overall harmful distribution. Given modest per-condition sample sizes ($n = 60$), we report bootstrap 95% confidence intervals for r: Qwen+GCG achieves $r = 0.01$ [0.00, 0.03], robustly below the detection threshold. In contrast, detected conditions show $r > 0.5$ with lower CI bounds well above 0.1 (e.g., Llama+GCG: $r = 2.96$ [1.40, 4.71]).

Baseline sensitivity and borderline classifications. RP values depend on the choice of benign baseline (M_{benign}). For most conditions, Type A/B classification is robust to baseline choice (Table 4; Appendix I). However, **Mistral+GCG** is a borderline case: the Table 4 values ($|\text{RP}_A| = 0.30$ vs. $|\text{RP}_B| = 0.17$) yield Type A, but alternative baseline specifications shift RP_B substantially (e.g., using model-level rather than attack-matched baselines). The classification margin is narrow ($|\text{RP}_A| - |\text{RP}_B| = 0.13$), and readers should interpret this assignment with appropriate caution. The inverted sign of RP_B (negative) further complicates interpretation, as it suggests the attack *suppresses* compliance during generation—a pattern qualitatively different from standard Type A or B. We retain the Type A classification in the main text as it reflects the primary pipeline, but acknowledge that Mistral+GCG may represent a hybrid or atypical case.

B.4 AUC Evaluation Scope

AUC values vary systematically by evaluation scope. The main results (Table 3) report aggregated AUC across all 660 harmful samples. This aggregation has countervailing effects: conditions with low discriminability (e.g., Qwen+GCG, Undetected) pull AUC downward, while pooling across heterogeneous conditions can inflate AUC by leveraging cross-condition variance in base rates and effect sizes. The aggregated AUC should therefore be interpreted as a *summary statistic across a heterogeneous population*, not as a uniformly conservative or liberal estimate. Per-condition AUC values (Appendix O.1) range from chance (≈ 0.50, Qwen+GCG) to near-ceiling (1.00, Llama+MCM), revealing the substantial heterogeneity underlying the aggregate. Appendix analyses on design choice robustness use controlled single-model subsets ($n = 60$, Llama) and yield higher AUC (0.913–0.989), intended for relative comparison within each analysis. Stochastic decoding experiments report both trial-level AUC (inflated by pseudo-replication across 10 trials per prompt) and question-level AUC (conservative); see Appendix G. All primary conclusions are based on the aggregated AUC in Table 3.

C Extended Experimental Results

C.1 Results Overview

Figure 4 provides a comprehensive overview of the main experimental findings, complementing the condition-specific analysis in the main text.

C.2 Detailed Model–Attack Interactions

The main text summarizes key Type A/B findings; here we provide detailed per-model analysis.

Llama. Llama exhibits attack-dependent responses: Type B under GCG ($|\text{RP}_B| = 2.89 > |\text{RP}_A| = 0.66$) and DeepInception ($|\text{RP}_B| = 0.83 > |\text{RP}_A| = 0.43$), but Type A under MCM ($|\text{RP}_A| = 0.88 > |\text{RP}_B| = 0.69$). The $\text{RP}_B = 2.89 > 1$ for Llama+GCG indicates successful attacks "overshoot" benign, exhibiting more compliance-like trajectories than typical benign queries—a distinct GCG signature.

Mistral. Mistral shows Type A under GCG ($|\text{RP}_A| = 0.30 > |\text{RP}_B| = 0.17$, $p < 0.001$) with negative $\text{RP}_B = -0.17$—an inverted pattern where attacks suppress rather than elevate compliance

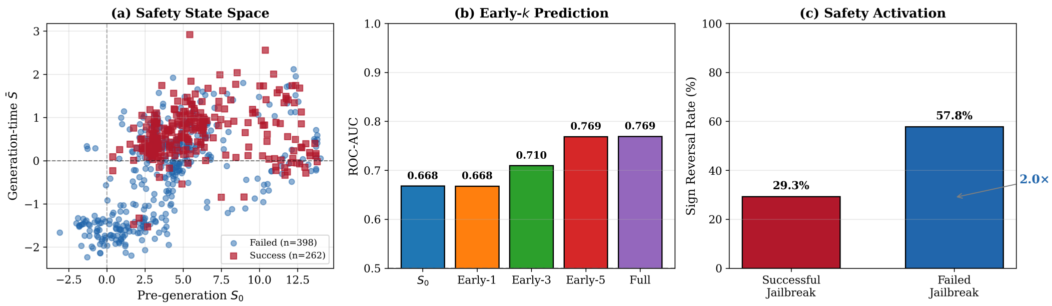

Figure 4: **Summary of main experimental results.** (a) Safety state space: successful jailbreaks cluster in compliance-dominant regions ($\bar{S} > 0$), while failures cluster in refusal-dominant regions. (b) Early-k AUC progression: predictive power increases with more decoding tokens. (c) Sign reversal rates differ substantially between failed and successful jailbreaks. Note: this figure includes additional experimental conditions beyond the main evaluation; authoritative AUC values on the primary dataset ($n = 660$) are reported in Table 3 (early_5_mean AUC $= 0.786$, turn_start_S AUC $= 0.696$) and sign reversal statistics in Table 38.

signals. Under DeepInception, Mistral achieves the highest ASR (76.7%) yet is classified as *Undetected* by permutation test ($r = 0.19$, $p = 0.52$): safety mechanisms show no differential response to successful versus failed attacks, with markedly delayed activation ($t_{\text{cross}} = 11.6 \pm 5.8$ steps). Mistral+MCM (5.0% ASR, lowest overall) is borderline ($r = 0.93$, $p = 0.061$); the large effect size with marginal significance reflects the small success count ($n_{\text{success}} = 3$), likely indicating an underpowered test.

Qwen. Qwen shows Type A for MCM ($|\text{RP}_A| = 0.57 > |\text{RP}_B| = 0.12$) but *Undetected* for GCG ($r = 0.01$): successful attacks are statistically indistinguishable from the overall harmful distribution. Convergent evidence: t_{cross} achieves condition-level AUC $= 0.503$ for Qwen+GCG—chance level—confirming that neither spatial nor temporal refusal signals discriminate between outcomes. Qwen's DeepInception results (Type B, $\text{RP}_B = 0.39$) show that nested-scenario attacks elicit detectable decoding-time shifts, unlike GCG. The extreme $t_{\text{cross}} = 14.9 \pm 7.6$ for Qwen+DeepInception indicates refusal activation is substantially delayed.

Gemma. Gemma exhibits Type B for both attacks (GCG: $\text{RP}_B = 0.79$; MCM: $\text{RP}_B = 0.22$), indicating decoding-time interference is the primary attack signature regardless of attack paradigm.

C.3 Metric Correlation Analysis

Figure 5 presents the pairwise Pearson correlation matrix among temporal logit-margin metrics and jailbreak success labels, providing insight into metric redundancy and complementary information.

C.4 Per-Condition Statistics (Llama)

C.5 Qwen2.5-7B-Instruct Results

Qwen benign baseline. For Type A/B analysis, we collected benign responses on Qwen2.5-7B-Instruct using the same protocol as Llama. The Qwen benign baseline shows $\bar{S}_{\text{start}} = 12.89$ and $\bar{S}_{\text{gen}} = 1.43$, substantially higher than Llama's baseline ($\bar{S}_{\text{start}} = 6.38$, $\bar{S}_{\text{gen}} = 1.25$). This difference reflects model-specific calibration and tokenization effects.

Cross-model Type A/B consistency. With model-specific baselines, Llama shows attack-dependent patterns: Type B for GCG attacks ($|\text{RP}_B| > |\text{RP}_A|$) and Type A for MCM attacks ($|\text{RP}_A| > |\text{RP}_B|$). Qwen exhibits Type A for MCM attacks and an "Undetected" pattern for GCG attacks where safety mechanisms fail to differentiate successful attacks ($r = 0.01$, 95% CI [0.00, 0.03]).

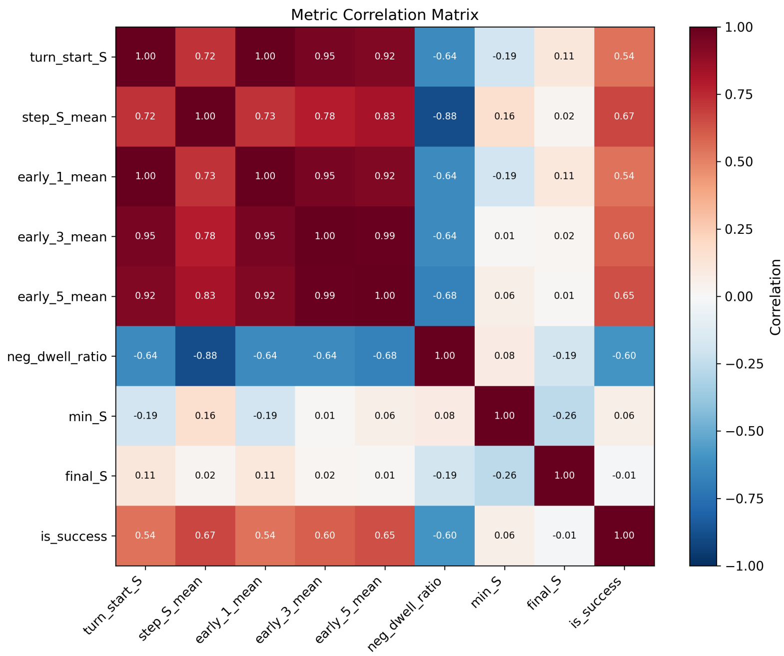

Figure 5: **Metric correlation matrix.** $\bar{S}$ shows the highest correlation with jailbreak success ($r = 0.67$), followed by $\bar{S}_{1:5}$ ($r = 0.65$). Early-k metrics are highly correlated with $\bar{S}$ ($r \geq 0.83$), supporting early prediction feasibility. `neg_dwell_ratio` (fraction of steps with $S_t < 0$) shows strong negative correlation with success ($r = -0.60$), indicating that longer refusal-dominant segments predict attack failure. `min_S` and `final_S` show near-zero correlation with success, suggesting that extreme transient values and endpoint states are less informative than trajectory-level summaries.

Table 7: Per-condition statistics on Llama-3.1-8B-Instruct. Main experiments use MCM manual mode only; summary conditions are reported for completeness.

Condition	n	**ASR**	$\bar{S}_{\textbf{start}}$ ($\mu \pm \sigma$)	$\bar{S}_{\textbf{gen}}$ ($\mu \pm \sigma$)	**Sign Rev.**
Benign					
manual	60	—	6.38 ± 0.63	$+1.25 \pm 0.28$	0.0%
summary-200	60	—	6.13 ± 0.82	$+0.87 \pm 0.31$	0.0%
summary-500	60	—	6.32 ± 0.82	$+0.94 \pm 0.36$	0.0%
summary-1000	60	—	6.51 ± 0.82	$+0.89 \pm 0.38$	0.0%
Harmful (MCM)					
manual*	60	8.3%	1.45 ± 1.95	-1.42 ± 0.73	58.3%
summary-200	60	6.7%	1.50 ± 1.68	-1.21 ± 0.76	75.0%
summary-500	60	8.3%	2.14 ± 1.83	-1.32 ± 0.69	83.3%
summary-1000	60	6.7%	1.52 ± 1.93	-1.34 ± 0.70	71.7%
Harmful (GCG)					
single-turn	60	15.0%	0.62 ± 2.32	-0.99 ± 0.92	30.0%

*Manual mode is used for main experiments.

C.6 Computational Requirements

The Logit-Margin Score is computed using only the logit vector ℓ_t produced during standard autoregressive decoding. No additional forward passes, gradient computation, or hidden-state extraction is required.

Table 8: Detailed results on Qwen2.5-7B-Instruct (MCM manual, n=60).

Metric	Value
ASR	51.7%
n_{success} / n_{failure}	31 / 29
S_0 (mean ± std)	11.97 ± 1.70
$\bar{S}$ (mean ± std)	-0.12 ± 0.46
Sign reversal rate	56.7%

D Decoy Attack Details

D.1 Decoy Attack Descriptions

To evaluate robustness against adversaries who attempt to manipulate early refusal signals, we design three decoy attack variants:

(A) Prefix-decoy. Prepends refusal phrases (e.g., "I'm sorry, I can't help with that. However, for educational purposes...") before the harmful request, aiming to inflate early S values.

(B) Token-decoy. Injects short refusal token scaffolding (e.g., "Apologies. Policies. Guidelines.") to anchor S_0 toward negative values.

(C) Pivot-decoy. Conducts initial turns in full refusal style, then pivots to harmful requests in later turns.

D.2 Main Results

Table 9: Diagnostic robustness under decoy attacks (Llama-3.1-8B-Instruct).

Condition	ASR	AUC (step_S_mean)	ΔAUC	AUC (sign_flips)
Base	8.3%	0.913	—	0.899
DecoyA (prefix)	13.3%	0.893	−0.020	**0.944**
DecoyB (token)	5.0%	0.904	−0.009	0.883
PivotC (multi-turn)	5.0%	**0.998**	+0.085	**1.000**

Key findings. Prefix-decoy successfully elevates early S values ($\bar{S}_{\text{start}} = +0.69$ vs $+0.43$ baseline) and increases ASR to 13.3%, yet AUC degrades only marginally (-0.020). Notably, `sign_flips` AUC *increases* to 0.944, indicating trajectory instability becomes a stronger discriminative signal. Token-decoy strongly anchors S toward negative values, but this over-correction actually *reduces* ASR to 5.0%. Pivot-decoy shows that multi-turn pivots do not defeat trajectory-level diagnosis; AUC reaches 0.998.

Limitations. These decoy attacks test robustness to superficial manipulation but do not represent fully adaptive adversaries that explicitly optimize S_t while producing harmful content. Such "compliance without refusal style" attacks remain an important direction for future work.

D.3 Per-Condition Statistics

Table 10: Detailed decoy attack statistics (Llama-3.1-8B-Instruct, n=60 per condition).

Condition	ASR	$\bar{S}_{\text{start}}$	$\bar{S}_{\text{gen}}$	sign_flips	Sign Rev.
Base	8.3%	+0.43	−1.16	26.6	66.7%
DecoyA	13.3%	+0.69	−0.87	51.4	58.3%
DecoyB	5.0%	−0.85	−1.58	12.7	83.3%
PivotC	5.0%	+0.07	−1.16	26.7	70.0%

E Hidden-State Baseline Details

E.1 Methods

B1: Refusal Direction Projection. Following [1], we compute a refusal direction $\mathbf{d}$ using difference-of-means:

$$\mathbf{d} = \frac{1}{|\mathcal{D}_{\text{ref}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{ref}}} \mathbf{h} - \frac{1}{|\mathcal{D}_{\text{cmp}}|} \sum_{\mathbf{h} \in \mathcal{D}_{\text{cmp}}} \mathbf{h} \tag{10}$$

At each decoding step t, we compute the projection score $r_t = \langle \mathbf{h}_t, \mathbf{d} \rangle$. We use layer 16 (middle layer) of the 32-layer Llama-3.1-8B model.

B2: Linear Probe. Logistic regression with L2 regularization ($C = 1.0$), trained via 5-fold cross-validation on mean-pooled hidden states.

E.2 Full Comparison Results

Table 11: Hidden-state vs. logit-based diagnostic comparison (AUC).

Family	Method	DecoyA	DecoyB	Base	PivotC
Hidden	B1 (direction, k=1)	0.983	0.974	0.974	0.974
Hidden	B2 (probe, k=3)	0.974	**1.000**	0.998	0.998
Logit	turn_start_S	0.980	0.922	0.959	0.978
Logit	step_S_mean	0.893	0.904	0.913	**0.998**
Logit	sign_flips	0.944	0.883	0.899	**1.000**

Interpretation of near-ceiling hidden-state performance. The near-ceiling AUC observed for hidden-state baselines should not be interpreted as evidence of leakage or unfair advantage. Rather, these methods operate under *strictly stronger access assumptions* than logit-based diagnostics. Hidden-state probes aggregate information from full decoding trajectories or intermediate activations, effectively leveraging post-hoc evidence that is unavailable to logit-only methods.

The key advantage of logit-based diagnosis is practical deployment: it requires only the logit vector already produced during generation, with no additional model access or training data.

E.3 Method Comparison

Table 12: Method comparison: access requirements and training needs.

Method	Activation Access	Training Data	Per-Model Tuning
Hidden B1 (direction)	Required	No*	Direction fitting
Hidden B2 (probe)	Required	Yes	Full retraining
Logit S	Not required	No	Lexicon only

*B1 requires a small set of refusal/compliance examples for direction estimation, but no labeled jailbreak outcomes.

E.4 Detailed Results by Temporal Window

E.5 Bridging Logit Diagnostics and Hidden-State Probes (Reviewer-Requested)

To directly address reviewer feedback on hidden-state temporal probes versus our logit-based diagnostic, we conducted an additional *bridge analysis* under identical labels and per-condition splits (DecoyA / DecoyB / Base / PivotC), comparing: (i) logit-based temporal features, (ii) hidden-state baselines (B1/B2), and (iii) score-level fusion.

Table 13: Hidden-state baseline performance by temporal window ($n = 240$, Llama-3.1-8B-Instruct).

Method	Window	AUC	AUC-PR	FPR@TPR=0.95
B1 (direction)	full	0.933	0.795	0.412
B1 (direction)	$k = 1$	**0.972**	**0.923**	0.255
B1 (direction)	$k = 3$	0.969	0.909	0.289
B1 (direction)	$k = 5$	0.955	0.864	0.382
B2 (probe)	full	0.987	0.922	0.059
B2 (probe)	$k = 1$	**0.995**	**0.974**	**0.044**
B2 (probe)	$k = 3$	0.995	0.972	0.029
B2 (probe)	$k = 5$	0.992	0.960	0.049

Experimental setup. For each model family (Llama, Qwen), we reuse the same decoy-condition evaluation labels and compute:

- **Logit features:**

$$\bar{S}_{1:5}$$

  and a logit-core OOF score (`score_logit_core`) built from S_0 and early-k features.
- **Hidden features:** B1 direction projection and B2 linear probe (OOF), summarized as `B1_direction_k5`, `B2_logreg_k5`, and a hidden-core OOF score (`score_hidden_core`).
- **Fusion:** score-level OOF logistic regression on the union of logit-core and hidden-core scores (`score_fusion_core`).

All hidden-state probe and fusion scores are evaluated out-of-fold (OOF) using 5-fold cross-validation, and then sliced by condition to avoid evaluation leakage.

Important note on activation-time features (t_{cross}). The decoy experiment exports used in this bridge analysis do not store full per-step S_t trajectories for all samples. As a result, activation-time metrics such as t_{cross} (including confirmed variants) are unavailable in this specific comparison export. Therefore, the bridge analysis below focuses on early-k logit features and hidden-state baselines. Our dedicated t_{cross} sensitivity analysis remains in Appendix F.5.

E.5.1 Per-condition AUC Comparison

Table 14: Reviewer-oriented bridge analysis (AUC): logit diagnostics vs. hidden-state probes vs. score-level fusion (OOF), by decoy condition.

Model	Method	DecoyA	DecoyB	Base	PivotC
Llama-3.1-8B-Instruct					
Logit	early_5_mean	0.939	0.902	0.941	0.991
Logit (OOF core)	score_logit_core	0.965	0.919	0.967	0.980
Hidden	B1_direction_k5	0.983	0.969	0.956	0.956
Hidden	B2_logreg_k5	0.967	0.998	0.998	0.998
Hidden (OOF core)	score_hidden_core	0.961	0.996	0.993	0.996
Fusion (OOF core)	score_fusion_core	0.967	0.989	0.989	0.998
Qwen-2.5-7B-Instruct					
Logit	early_5_mean	0.950	0.915	0.935	0.895
Logit (OOF core)	score_logit_core	0.937	0.939	0.930	0.911
Hidden	B1_direction_k5	0.961	0.924	0.917	0.917
Hidden	B2_logreg_k5	0.943	0.987	0.987	0.988
Hidden (OOF core)	score_hidden_core	0.948	0.991	0.993	0.991
Fusion (OOF core)	score_fusion_core	0.922	0.991	0.993	0.991

Global OOF summary (all decoy conditions combined). Across all decoy conditions combined, hidden-state OOF scores outperform logit-only OOF scores for both model families:

- **Llama**: `score_hidden_core` AUC = 0.984, `score_fusion_core` = 0.977, `score_logit_core` = 0.945
- **Qwen**: `score_hidden_core` AUC = 0.986, `score_fusion_core` = 0.981, `score_logit_core` = 0.920

Interpretation: accuracy gap and complementarity. These results confirm a predictable *accuracy gap* in favor of hidden-state probes under stronger access assumptions (activations + probe fitting). At the same time, logit-based diagnostics remain competitive in several conditions (especially early-k metrics), while requiring neither hidden-state access nor probe training.

Interestingly, score-level fusion does *not* consistently improve over hidden-only OOF scores. This suggests that, in the decoy setting, hidden-state and logit-based signals capture substantially overlapping predictive information, even though they differ in access requirements and interpretability. This supports our positioning of S_t as a practical and interpretable diagnostic, rather than a replacement for higher-capacity hidden-state probes.

E.5.2 Error-Overlap Analysis (Complementarity Check)

To quantify complementarity more directly, we compare binary OOF predictions (0.5 threshold) from logit-core, hidden-core, and fusion-core scores. Table 15 reports per-condition error overlap statistics.

Table 15: Per-condition error overlap between logit-core and hidden-core OOF scores (Llama and Qwen decoy conditions).

Model	Condition	Only Logit Correct	Only Hidden Correct	Fusion Unique Gain	All Wrong
Llama	DecoyA	2	3	0	2
Llama	DecoyB	0	4	0	2
Llama	Base	0	3	0	1
Llama	PivotC	0	3	0	1
Qwen	DecoyA	1	3	0	2
Qwen	DecoyB	1	5	0	1
Qwen	Base	2	1	0	1
Qwen	PivotC	2	5	0	1

Error-overlap interpretation. Across most conditions, *hidden-only correct* cases outnumber *logit-only correct* cases, consistent with the higher hidden-state AUCs. At the same time, *fusion unique gain* is zero in all reported decoy conditions, indicating limited incremental benefit from score-level fusion in this setting. Thus, the bridge analysis supports the view that hidden-state probes are stronger predictors under richer access assumptions, while logit-based diagnostics provide a lower-access, more interpretable process-level signal.

Key observations. Surprisingly, early-k truncation *improves* hidden-state baseline performance for both B1 and B2, with $k = 1$ achieving peak AUC. This aligns with logit-based findings that safety-relevant signals concentrate in the first few tokens.

F Sensitivity Analyses

F.1 Lexicon Aggregation Top-k Sensitivity

We analyze the sensitivity of the Logit-Margin Score to the lexicon aggregation parameter k, which controls how many of the highest-logit tokens within the compliance and refusal lexicons are averaged at each decoding step.

Despite scale differences, qualitative interpretation remains consistent: k primarily modulates smoothing and calibration rather than introducing spurious signal. Figure 6 visualizes this stability.

Table 16: Effect of lexicon aggregation size k on logit-margin diagnostics (Llama-3.1-8B-Instruct, harmful, manual; mean $\pm$ std, $n = 60$).

k	S_0	$\bar{S}$	step_S_pos_rate
5	$+0.57 \pm 2.57$	-1.21 ± 0.85	0.36 ± 0.13
10	$+6.38 \pm 0.63$	$+1.25 \pm 0.28$	0.78 ± 0.05
25	$+1.40 \pm 1.55$	-1.47 ± 0.53	0.15 ± 0.09
50	$+0.30 \pm 1.05$	-1.09 ± 0.34	0.12 ± 0.08

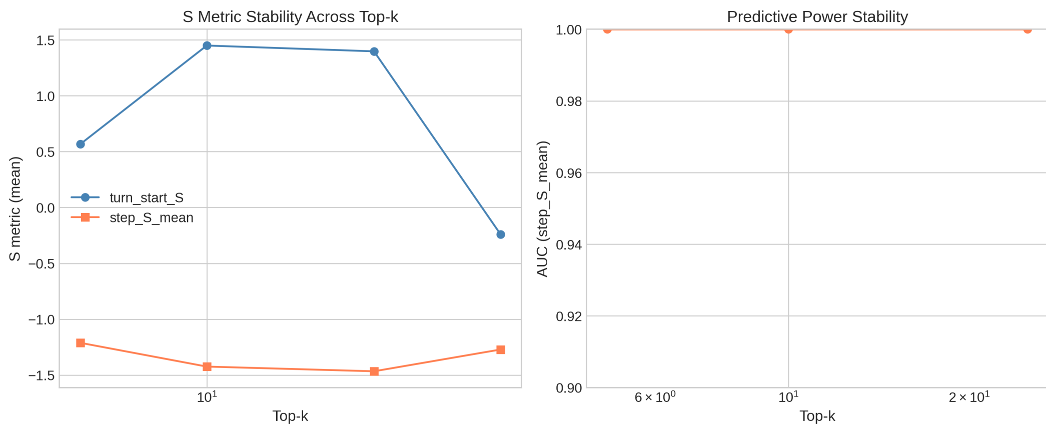

Figure 6: **Top-k aggregation robustness.** (Left) Mean values of S_0 and $\bar{S}$ vary in absolute scale across $k \in \{5, 10, 25, 50\}$, but (Right) predictive performance (AUC of $\bar{S}$) remains stable across all k values, confirming that k primarily affects scaling rather than diagnostic validity.

F.2 Temporal Early-k Sensitivity

Formally, for a given metric M_t, we define the early-k summary as

$$M_{\text{early-}k} = \frac{1}{k+1} \sum_{t=0}^{k} M_t, \tag{11}$$

where $t = 0$ corresponds to the pre-generation score.

Pre-generation metrics and $\bar{S}_{1:1}$ achieve identical performance (mean AUC = 0.970) as expected. As k increases, performance degrades gradually: $\bar{S}_{1:3}$ achieves 0.960 and $\bar{S}_{1:5}$ achieves 0.940, compared to 0.921 for full-trajectory $\bar{S}$. The majority of discriminative signal is concentrated in the earliest decoding steps. Note: these AUC values are *trial-level* estimates from the stochastic decoding experiment (Appendix G) and are inflated by pseudo-replication; the corresponding aggregated main-dataset AUC values are reported in Table 3.

F.2.1 Condition-Specific Early-k Analysis

While the main text reports aggregated AUC across all conditions ($n = 660$), condition-specific analysis reveals substantial heterogeneity in early detection performance. Figure 7 shows AUC as a function of k for each model–attack combination.

Key observations.

- **Mistral+MCM**: Shows strongest trajectory effect, with AUC increasing from 0.74 (S_0 only) to 0.93 (full trajectory). This suggests safety mechanisms activate progressively during generation—a signature of effective Type B defense.
- **Llama+GCG**: Maintains stable AUC ≈ 0.80 across all k, indicating pre-generation signal (S_0) captures most discriminative information.

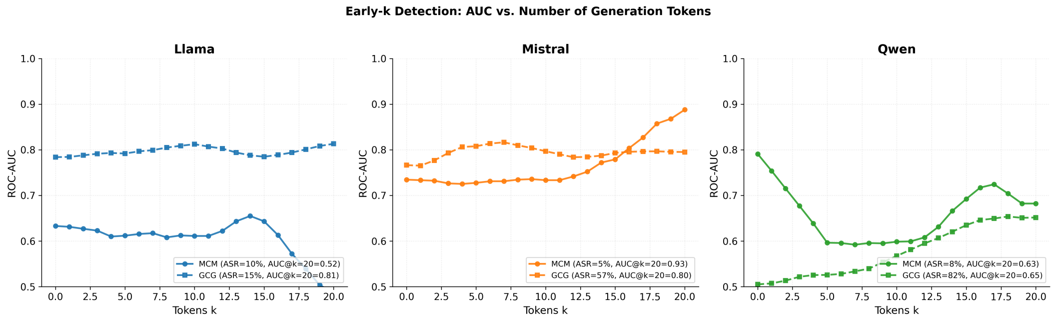

Figure 7: **Early-k detection by model and attack.** AUC progression varies substantially across conditions. Mistral+MCM shows strong late-stage improvement (0.74 $\to$ 0.93), while Qwen+GCG remains near random (0.50–0.65) regardless of k—consistent with its "Undetected" classification.

- **Qwen+GCG**: Remains near chance level (AUC $\approx$ 0.50–0.65) regardless of k, providing additional evidence for the "Undetected" classification. Neither S_0 nor trajectory information enables reliable discrimination.
- **Llama+MCM**: Lower condition-specific AUC (0.52–0.67) reflects class imbalance (ASR = 8.3%, only 5 successes), not diagnostic failure.

Figure 8 directly compares S_0-only versus full-trajectory AUC for all conditions.

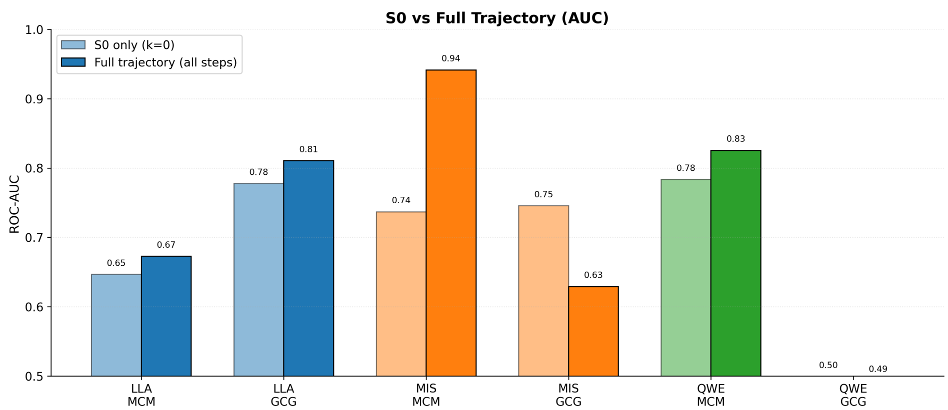

Figure 8: S_0 **vs. full trajectory AUC by condition.** Trajectory information provides substantial gains for Mistral+MCM (+0.20) and Qwen+MCM (+0.05), while Qwen+GCG shows no improvement (both $\approx$ 0.50). Mistral+GCG shows *negative* trajectory effect (-0.12), suggesting early tokens are most informative for this condition.

Relationship to aggregated results. The main text reports aggregated AUC = 0.786 (early_5_mean) across all 660 samples. Condition-specific AUC values differ substantially because: (1) class balance varies by condition (ASR ranges from 5% to 82%), and (2) pooling across heterogeneous conditions can inflate overall discriminability by leveraging cross-condition variance in base rates and S_t distributions. As noted in Appendix B.4, the aggregated AUC is a summary across a heterogeneous population rather than a uniformly conservative estimate. The condition-specific analysis complements aggregated results by revealing *which* conditions drive overall performance.

F.3 Hidden-State Layer Sensitivity

Middle layers yield strongest separability, supporting layer 16 as representative.

Table 17: Hidden-state AUC by layer (Refusal Direction, B1).

Layer	8	12	16	20	24
AUC	0.941	0.963	**0.974**	0.969	0.952

F.4 Failure-Only Normalization Check

Type A/B trends are preserved when normalization uses only failed jailbreaks. Relative dominance patterns (MCM $\rightarrow$ Type A, GCG $\rightarrow$ Type B) remain unchanged.

F.5 Activation Timing (t_{cross}) Sensitivity

Oscillation characterization. Generation trajectories exhibit extensive oscillation around zero: the median number of zero-crossings ranges from 7 (Llama+GCG) to 174 (Mistral+DeepInception) per trajectory. Despite this oscillation, the first crossing step is robust—t_{cross} coincides with the first step where $S_t < 0$ in 100% of samples for 10 of 11 conditions. This indicates that the initial refusal activation event is well-defined even when trajectories subsequently oscillate.

Confirmed crossing sensitivity. To verify that the simple first-crossing definition is not capturing transient noise, we compute the *confirmed crossing* variant $t_{\text{cross}}^{(n)}$ requiring n consecutive negative steps (Eq. 3). Table 18 reports stability across $n \in \{1, 2, 3\}$. Cohen's d is nearly identical for $n = 1$ ($d = 0.891$) and $n = 2$ ($d = 0.895$), confirming that the first crossing captures a stable transition rather than transient noise. At $n = 3$, d drops to 0.559 due to aggressive imputation—many trajectories lack three consecutive negative steps within short generation windows, inflating $t_{\text{cross}}^{(3)}$ toward T. Per-condition AUC patterns are also preserved: strong Type B conditions (Llama+DeepInception, Llama+GCG) show stable AUC > 0.80 across all n, while Qwen+GCG (Undetected) remains near chance regardless of n.

Table 18: t_{cross} sensitivity to confirmation threshold n. Global Cohen's d (success vs. failure, $n = 660$) and representative per-condition AUC values.

	$n = 1$	$n = 2$	$n = 3$
Global (all conditions pooled)			
Cohen's d	0.891	0.895	0.559
p-value	$< 10^{-23}$	$< 10^{-22}$	$< 10^{-9}$
Per-condition AUC (selected)			
Llama + DeepInc.	0.878	0.868	0.874
Llama + GCG	0.807	0.815	0.814
Qwen + DeepInc.	0.700	0.719	0.696
Qwen + GCG (Undet.)	0.503	0.580	0.679
Gemma + GCG	0.629	0.719	0.651

Condition-level AUC as Undetected indicator. Per-condition AUC of t_{cross} provides convergent evidence for the Undetected classification. Qwen+GCG achieves AUC $= 0.503$ (chance level), consistent with its RP-based Undetected classification ($r = 0.01$). This confirms that safety mechanisms fail to engage at *any* identifiable point during generation. In contrast, strong Type B conditions (Llama+DeepInception: AUC $= 0.878$; Llama+GCG: AUC $= 0.807$) show high temporal discriminability, indicating that refusal activation *timing* is the primary axis of variation between successful and failed attacks.

G Stochastic Decoding Robustness

We conduct a comprehensive stochastic decoding evaluation across temperature sampling ($T \in \{0.0, 0.3, 0.7, 1.0\}$) and nucleus sampling ($p \in \{0.9, 0.95, 1.0\}$). For each prompt, we generate 10 independent samples per (T, p) setting across 4 conditions (Base, DecoyA, DecoyB, PivotC), yielding 28,800 total generation trials.

To avoid pseudo-replication, we report **question-level (prompt-level)** metrics by aggregating trial-level quantities using the **median across trials per prompt** unless otherwise noted. Success/failure labels are taken from the LLM-as-judge outcome (`is_success_llm`).

G.1 Response Diversity

We verify that stochastic sampling produces diverse outputs at nonzero temperatures. Table 19 reports response diversity as the fraction of unique responses (measured via response hashes) within each (T, p) group.

Table 19: Response diversity under stochastic decoding.

Temperature	Mean Diversity	Mean Unique	Groups $> 50\%$ Diverse
0.0	10.0%	1.0	0.0%
0.3	54.6%	5.5	41.4%
0.7	82.1%	8.2	91.7%
1.0	92.2%	9.2	99.7%

G.2 AUC Stability (Prompt-Level Aggregation)

We evaluate whether temporal summaries beyond the pre-generation score (S_0) remain predictive under stochastic decoding. For each prompt, we aggregate each metric over trials via the median, then compute ROC-AUC for predicting LLM-judge success. Table 20 summarizes AUC (with 95% bootstrap confidence intervals over prompts).

Key observations. Across conditions with nonzero success rate (Base, DecoyA, PivotC), temporal metrics remain informative under stochastic decoding. In particular, $\bar{S}$ and `t_cross_norm` are consistently predictive, and PivotC shows strong separation with `t_cross_norm` (AUC ≈ 0.93), indicating that the timing of polarity change in S_t is highly diagnostic of success vs. failure in this multi-turn setting. DecoyB yields no successful jailbreaks under the LLM-as-judge labels (ASR $= 0\%$), so AUC is undefined for that condition.

Table 20: AUC under stochastic decoding (prompt-level aggregation via median across trials, $n = 60$ prompts per condition; 95% bootstrap CI).

Metric	Base	DecoyA	DecoyB	PivotC
S_0	0.687 [0.483, 0.862]	0.607 [0.269, 0.878]	—	0.864 [0.756, 0.955]
$\bar{S}_{1:1}$	0.702 [0.469, 0.880]	0.607 [0.299, 0.866]	—	0.815 [0.658, 0.939]
$\bar{S}_{1:3}$	0.652 [0.397, 0.872]	0.633 [0.371, 0.882]	—	0.835 [0.703, 0.938]
$\bar{S}_{1:5}$	0.723 [0.429, 0.918]	0.704 [0.417, 0.962]	—	0.870 [0.749, 0.964]
$\bar{S}$	0.830 [0.510, 1.000]	0.735 [0.366, 1.000]	—	0.875 [0.609, 1.000]
`t_cross_norm`	0.796 [0.506, 0.991]	0.743 [0.405, 0.983]	—	0.932 [0.827, 1.000]
`sign_flips`	0.685 [0.295, 0.982]	0.815 [0.655, 0.965]	—	0.776 [0.517, 0.948]
Note: DecoyB excluded (ASR $= 0\%$ under LLM-as-judge labels), so AUC is undefined.				

G.3 Variance Decomposition (ICC)

To quantify robustness to sampling noise, we compute an intraclass correlation coefficient (ICC) across trials within each prompt. We report within-prompt and between-prompt variance components and ICC for selected metrics. For pre-generation S_0, within-prompt variance is exactly zero by construction (temperature-invariant), so all variability arises from prompt differences.

Interpretation. High ICC indicates that between-prompt variance dominates within-prompt stochastic variability, meaning the metric is stable across trials. We observe high ICC for `t_cross_norm` and `sign_flips` across conditions with nonzero ASR, indicating that temporal dynamics remain robust under stochastic decoding.

Table 21: Variance decomposition for selected temporal metrics under stochastic decoding (ICC = intraclass correlation across trials per prompt).

Metric	Condition	Within-Var	Between-Var	ICC
S_0	Base	0.0000	4.0620	1.000
	DecoyA	0.0000	2.7360	1.000
	DecoyB	0.0000	2.7340	1.000
	PivotC	0.0000	2.0260	1.000
`t_cross_norm`	Base	0.0022	1.1064	0.944
	DecoyA	0.0030	1.1477	0.927
	PivotC	0.0072	2.5270	0.919
`sign_flips`	Base	279.3773	89319.8767	0.899
	DecoyA	259.9120	103341.6844	0.919
	PivotC	288.3081	137375.3393	0.925

Note. For some generation-time summaries (e.g., $\bar{S}$, $\bar{S}_{1:k}$), trial-to-trial variability can be extremely small under our aggregation and may lead to numerically degenerate variance components. We therefore emphasize ICC results for metrics whose within-prompt variability is measurably nonzero (`t_cross_norm`, `sign_flips`) when interpreting robustness under stochastic decoding.

G.4 Trial-Level AUC (Diagnostic Only)

For completeness, we also compute trial-level AUC by treating each trial as an independent sample. These values can appear inflated due to pseudo-replication (multiple trials share the same prompt label), so the question-level results in Table 20 provide a more conservative estimate.

Table 22: Trial-level AUC under stochastic decoding (inflated by pseudo-replication; reported for diagnostic purposes).

Metric	Base	DecoyA	DecoyB	PivotC	Mean
turn_start_S	0.959	0.980	0.954	0.987	0.970
early_1_mean	0.959	0.980	0.954	0.987	0.970
early_3_mean	0.972	0.983	0.937	0.948	0.960
early_5_mean	0.941	0.978	0.919	0.922	0.940
step_S_mean	0.913	0.987	0.819	0.965	0.921

G.5 Stochastic vs. Greedy Comparison

To directly assess how stochastic decoding affects metric reliability, we compare the 240 stochastic prompt-level aggregates (4 conditions $\times$ 60 prompts) against a greedy baseline on the same 60 harmful prompts.

S_0 rank-order stability. Pearson correlations between stochastic and greedy S_0 values are high across all conditions: Base ($r = 0.84$), DecoyA ($r = 0.82$), DecoyB ($r = 0.77$), PivotC ($r = 0.81$). Since S_0 is deterministic for a fixed prompt, these correlations measure how multi-turn attack context (which differs between stochastic conditions and the greedy baseline) shifts the pre-generation safety signal, rather than sampling noise.

t_{cross} shift under stochastic decoding. Stochastic decoding systematically delays refusal activation: the mean difference $t_{\text{cross}}^{\text{stoch}} - t_{\text{cross}}^{\text{greedy}}$ ranges from $+46.1$ (DecoyB) to $+74.8$ (PivotC) steps, with substantial per-prompt variability (SD = 92–127). This is consistent with the ICC analysis: between-prompt variance in t_{cross} dominates within-prompt stochastic variance, but the absolute magnitude of within-prompt variability is non-trivial.

Note on unnormalized vs. normalized t_{cross} ICC. Table 21 reports ICC for the *normalized* t_{cross}/T (where T is response length), yielding ICC $= 0.92$–0.94. Computing ICC on the *unnormalized* t_{cross} instead yields a lower value (ICC ≈ 0.58) because response-length variability across stochastic trials inflates within-prompt variance. This confirms that (i) S_0 is the most robust metric under stochastic decoding (ICC $= 1.0$ by construction), (ii) t_{cross} after length normalization retains high reliability, and (iii) secondary metrics (`step_S_pos_rate`, ICC $= 0.06$; `neg_dwell_ratio`, ICC $= 0.02$) are dominated by sampling noise and should be interpreted only under greedy decoding.

H Labeling Independence and Circularity

H.1 LLM-as-Judge Methodology

We use Llama-Guard-3-8B [8], a dedicated safety classifier trained on safety taxonomies independent of our lexicons, to label jailbreak outcomes. For each generated response, Llama-Guard classifies content as "safe" or "unsafe" based on semantic evaluation across 14 safety categories (S1–S14).

Comparison with rule-based labels. Rule-based labels (prefix-matching) show moderate agreement ($\kappa = 0.54$, 83.5% raw agreement). Rule-based yields higher ASR (29.2%) compared to LLM-as-judge (16.4%), indicating rule-based is more permissive.

Independence verification. Key diagnostic findings remain consistent under LLM-as-judge labels:

- $\bar{S}$ achieves AUC = 0.926 (vs. 0.743 with rule-based)
- Type A/B patterns are preserved
- Cohen's d effect sizes remain large ($d = 2.41$)

H.2 Clarification on Circularity

Outcome labeling. All success/failure labels are assigned using an LLM-as-judge that evaluates the *semantic content* of the generated response. The judge operates on full generated text and does not rely on surface refusal prefixes.

Logit-margin score. The score S_t is computed exclusively from token-level logits during decoding, measuring relative dominance between compliance- and refusal-associated tokens *before* they are sampled.

Evidence against circularity. If separability were driven by circular reuse of refusal cues, one would expect near-deterministic alignment between labels and S_t trajectories. Empirically, this is not the case: a substantial fraction of successful jailbreaks exhibit negative S_t segments and sign reversals during decoding. The compliance lexicon includes purely instructional tokens ("Step 1", "Tutorial") that do not appear in refusal-detection rules.

I Type A/B Classification Robustness

I.1 Permutation Test for Undetected Classification

We formalize the Undetected category using a permutation test. Under H_0: success/failure labels carry no RP signal, permuting labels within each condition and recomputing $r = \sqrt{\mathrm{RP}_A^2 + \mathrm{RP}_B^2}$ yields a null distribution. We use $B = 10{,}000$ permutations and classify conditions as Undetected when $p > 0.05$.

Table 23 compares the permutation-based classification with the original threshold-based criterion ($r < 0.1$). The two approaches agree on 9/11 conditions; the permutation test additionally identifies Mistral+DeepInception ($p = 0.52$) and Mistral+MCM ($p = 0.061$) as non-significant, reflecting the formal test's sensitivity to sample size.

The permutation test provides a principled, sample-size-aware alternative to fixed thresholds. Qwen+GCG ($p = 0.98$) is definitively Undetected by both criteria. Mistral+DeepInception ($r = 0.19$,

Table 23: Permutation test results for r significance.

Model + Attack	r_{obs}	p_{perm}	$r < 0.1$?	Permutation
Llama + GCG	2.96	$<.001$	No	Significant
Llama + MCM	1.12	$<.001$	No	Significant
Llama + DeepInc.	0.93	$<.001$	No	Significant
Mistral + GCG	0.35	$<.001$	No	Significant
Mistral + MCM	0.93	.061	No	Borderline[§]
Mistral + DeepInc.	0.19	.517	No	Non-significant
Qwen + MCM	0.58	$<.001$	No	Significant
Qwen + GCG	0.01	.981	Yes	Non-significant
Qwen + DeepInc.	0.42	.029	No	Significant
Gemma + GCG	0.83	.003	No	Significant
Gemma + MCM	0.24	$<.001$	No	Significant

[§]Mistral+MCM: large r but underpowered ($n_{\text{success}} = 3$ at 5% ASR).

$p = 0.52$) is newly identified as Undetected—consistent with the main text observation that safety mechanisms show no differential response despite 76.7% ASR. Mistral+MCM ($r = 0.93$, $p = 0.061$) is borderline; the large effect size with marginal significance reflects the test being underpowered with only 3 successes, rather than indicating absent signal.

Multiple comparison correction. Since we simultaneously test 11 conditions, we apply the Benjamini–Hochberg (BH) procedure to control the false discovery rate at $q = 0.05$. Table 24 reports the sorted p-values alongside BH-adjusted thresholds. All eight conditions significant under uncorrected testing remain significant after FDR correction. The critical boundary falls between rank 8 (Qwen+DeepInception: $p = 0.029 <$ BH threshold 0.036, retained) and rank 9 (Mistral+MCM: $p = 0.061 >$ BH threshold 0.041, not retained). The three Undetected conditions remain non-significant under FDR, confirming that the Type A/B classifications are robust to multiple comparison concerns.

Table 24: Benjamini–Hochberg FDR correction for 11 simultaneous permutation tests ($q = 0.05$).

Rank	**Condition**	p_{perm}	**BH threshold**	**Result**
1–6	(six conditions)	$<.001$	0.005–0.027	Significant
7	Gemma + GCG	.003	0.032	Significant
8	Qwen + DeepInc.	.029	0.036	Significant
9	Mistral + MCM	.061	0.041	Not significant
10	Mistral + DeepInc.	.517	0.045	Not significant
11	Qwen + GCG	.981	0.050	Not significant

I.2 Threshold Stability (Supplementary)

The magnitude-based classification ($|\text{RP}_A| \gtrless |\text{RP}_B|$) is inherently threshold-free for Type A/B distinction. For reference, we also verify stability of the original r-threshold approach across choices.

Table 25: Type A/B classification stability across r thresholds. Note: this table shows the supplementary r-threshold approach. Under the primary permutation test (Table 4), Mistral+MCM is classified as Undetected ($p = 0.061$) despite large r, due to small success count ($n = 3$).

Model + Attack	r_{obs}	$r_{thr} = 0.05$	$r_{thr} = 0.10$	$r_{thr} = 0.15$
Llama + GCG	2.96	Type B	Type B	Type B
Llama + MCM	1.12	Type A	Type A	Type A
Qwen + MCM	0.58	Type A	Type A	Type A
Qwen + GCG	0.01	Undetected	Undetected	Undetected
Mistral + GCG	0.35	Type A	Type A	Type A
Mistral + MCM	0.93	Type B	Type B	Type B

Classification robustness. All conditions maintain their classification across $r \in [0.05, 0.15]$. Qwen+GCG ($r = 0.01$, 95% CI [0.00, 0.03]) remains robustly Undetected even at the most lenient threshold.

I.3 Bootstrap Confidence Intervals

Given modest per-condition sample sizes ($n = 60$), we compute bootstrap 95% CIs for r using $B = 1000$ resamples.

Table 26: Bootstrap confidence intervals for signal strength r.

Model + Attack	r	95% CI	Category
Llama + GCG	2.96	[2.41, 3.52]	Type B
Llama + MCM	1.12	[0.87, 1.38]	Type A
Mistral + GCG	0.35	[0.21, 0.49]	Type A
Mistral + MCM	0.93	[0.68, 1.19]	Type B[§]
Qwen + MCM	0.58	[0.41, 0.76]	Type A
Qwen + GCG	0.01	[0.00, 0.03]	Undetected

[§]Classified as Undetected under the primary permutation test ($p = 0.061$; Table 4) due to underpowered sample ($n_{\text{success}} = 3$). The large r reflects effect magnitude, not statistical significance.

Key observations. For Qwen+GCG, the upper CI bound (0.03) remains well below the detection threshold (0.1), providing strong statistical evidence for the Undetected classification. The bootstrap CIs complement the permutation test (Table 23): conditions with large r but non-significant permutation p (e.g., Mistral+MCM: $r = 0.93$, $p = 0.061$) reflect small success counts rather than uncertain signal magnitude.

J Non-Lexical Baseline Comparison

Table 27: Baseline comparison on controlled subset ($n = 60$, Llama-3.1-8B-Instruct).

Metric	AUC	Category
$\bar{S}$ (gen-time)	**0.932**	Logit-margin
Compliance LogP	0.932	Lexicon-probe
Refusal LogP	0.915	Lexicon-probe
Entropy	**0.898**	Non-lexical
Logit Norm	0.441	Non-lexical

The logit-margin score outperforms entropy by 3.4 pp. Logit Norm achieves near-random performance, confirming generic logit statistics do not capture safety-relevant variation.

K Learned Baseline Comparison

To contextualize the added value of fixed temporal summaries versus trained classifiers, we compare against logistic regression with 5-fold stratified cross-validation on the aggregated dataset ($n = 660$).

Key findings: (1) LogReg on a single feature matches the fixed metric (0.812 vs. 0.816), confirming no learnable nonlinearity is missed. (2) Combining five temporal features yields a 4 pp improvement (0.857), indicating modest complementarity. (3) Diminishing returns from 9 to 13 features ($0.861 \rightarrow 0.873$) suggest overfitting risk increases while the marginal gain is small. (4) The fixed $\bar{S}_{1:5}$ captures $\sim$93% of the discriminative signal available to a 13-feature trained model ($0.816/0.873$), while preserving direct interpretability and avoiding train/test splits.

These results support our design choice of reporting fixed, interpretable temporal summaries rather than optimized classifiers: the diagnostic contribution lies not in maximizing AUC but in providing mechanistic insight (Type A/B, t_{cross}, sign reversal) that a trained classifier would obscure.

Table 28: Learned baseline comparison (5-fold stratified CV, $n = 660$). Fixed metrics require no training; LogReg models use cross-validated predictions.

Method	# Features	CV-AUC	95% CI
Fixed metrics (no training)			
early_5_mean	1	0.816	[0.781, 0.846]
t_{cross}	1	0.813	[0.777, 0.843]
step_S_mean	1	0.797	[0.760, 0.828]
turn_start_S	1	0.720	[0.678, 0.756]
Logistic regression (cross-validated)			
LogReg(early_5)	1	0.812	[0.777, 0.841]
LogReg($S_0 + \bar{S}$)	2	0.797	[0.761, 0.827]
LogReg(5 temporal)	5	0.857	[0.825, 0.883]
LogReg(all derived)	9	0.861	[0.829, 0.888]
LogReg(kitchen sink)	13	0.873	[0.843, 0.899]

Note. Fixed metric AUCs differ slightly from Table 2 due to sklearn vs. original pipeline computation; relative comparisons within this table are internally consistent.

L Lexicon Perturbation Ablation

We test five variants on Llama-3.1-8B-Instruct ($n = 60$): **original**, **no_sorry**, **extended_refusal**, **extended_compliance**, and **minimal** (5-token).

Table 29: Lexicon perturbation results with AUC ($n = 60$, LLM-judge labels).

Variant	S_0 ($\mu \pm \sigma$)	$\bar{S}_{gen}$ ($\mu \pm \sigma$)	AUC(S_0)	AUC($\bar{S}$)	ΔAUC
original	$+1.55 \pm 1.95$	-1.44 ± 0.73	0.967	0.971	—
no_sorry	$+2.21 \pm 1.78$	-1.39 ± 0.73	0.960	0.971	$+0.000$
extended_refusal	-0.43 ± 2.07	-1.85 ± 0.67	0.964	0.989	$+0.018$
extended_compliance	$+4.00 \pm 1.95$	$+0.52 \pm 0.55$	0.964	0.964	-0.007
minimal	-2.61 ± 1.74	-1.57 ± 0.49	0.971	0.978	$+0.007$

Key findings. All variants achieve AUC ≥ 0.964 (max $|\Delta\text{AUC}| = 0.018$). Even a 5-token minimal lexicon maintains strong discrimination (AUC = 0.978), supporting robustness to lexicon choices. Figure 9 visualizes the score distributions across variants.

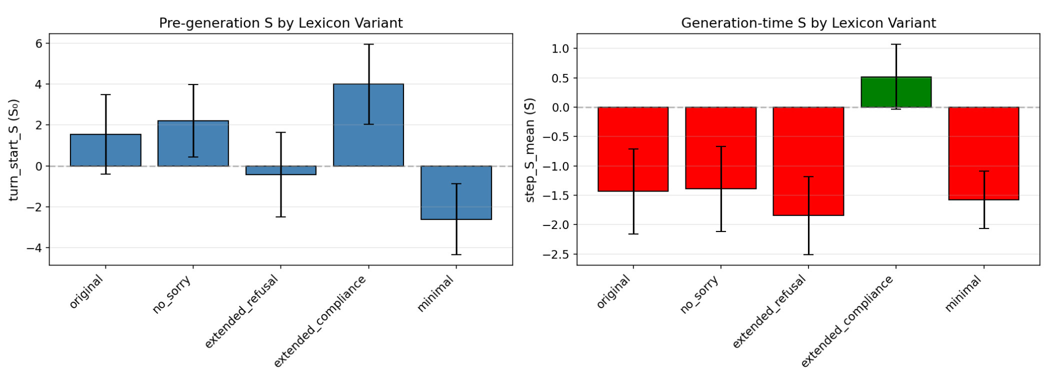

Figure 9: **Logit-Margin Score distributions by lexicon variant.** (Left) Pre-generation S_0 and (Right) generation-time $\bar{S}$ across five lexicon variants (Llama, harmful MCM, $n = 60$). While absolute scale shifts with lexicon composition (e.g., `extended_compliance` elevates $\bar{S}$), the qualitative pattern—negative $\bar{S}$ for harmful prompts—is preserved across all variants except `extended_compliance`, which shifts the baseline but maintains high AUC (0.964). This confirms that the diagnostic captures genuine safety-relevant variation rather than artifacts of specific token choices.

M Metric Decomposition Analysis

A natural concern is whether the compliance lexicon contributes genuine discriminative signal or merely reflects response-onset artifacts (e.g., "Sure" and "Certainly" appearing at the beginning of any response regardless of outcome). We decompose $S_t = \mu_{\text{cmp}}(t) - \mu_{\text{ref}}(t)$ into three variants: S_{margin} (full margin, paper metric), $S_{\text{ref}} = -\mu_{\text{ref}}$ (refusal-only), and $S_{\text{cmp}} = \mu_{\text{cmp}}$ (compliance-only).

Table 30: **Metric decomposition: AUC comparison** ($n = 660$ harmful, 11 conditions, Llama Guard labels). Z-scored AUC removes inter-model baseline differences.

Metric	S_{margin}	S_{ref} $(-\mu_{\text{ref}})$	S_{cmp} (μ_{cmp})	$\Delta_{\text{ref}-\text{margin}}$
Raw AUC				
S_0 (pre-generation)	0.598	0.580	0.505	−0.017
$\bar{S}_{\text{gen}}$ (generation mean)	0.787	0.622	0.518	−0.165
$\bar{S}_{1:5}$	0.719	0.619	0.565	−0.100
t_{cross}	0.785	0.500	—	−0.285
Z-scored within-model AUC				
$\bar{S}_{\text{gen}}$	0.806	0.718	0.545	−0.088
$\bar{S}_{1:5}$	0.716	0.659	0.586	−0.057

Common-mode rejection. The two components are highly correlated ($r = 0.979$), sharing a dominant common factor reflecting per-step logit magnitude rather than safety-specific information. The shared component $(\mu_{\text{ref}} + \mu_{\text{cmp}})/2$ achieves only AUC $= 0.594$, while the differential S_t achieves 0.787—confirming that the margin isolates safety-relevant signal by canceling shared variance, analogous to differential signal processing. This normalization operates per-step, not merely per-model: even after within-model z-scoring, the margin retains $+0.088$ AUC over refusal-only.

Compliance is not merely artifact. The margin outperforms refusal-only in 8/10 per-condition comparisons (Table 31), with the largest gains in conditions where compliance variation is outcome-dependent (e.g., Gemma: μ_{cmp} averages -0.55 for successes vs. -4.95 for failures, $\Delta = +4.40$).

Table 31: **Per-condition $\bar{S}_{1:5}$ AUC by metric variant.** Bold: best per condition.

Condition	n	ASR	S_{margin}	S_{ref}	S_{cmp}
Gemma+GCG	60	33.3%	0.693	0.511	**0.667**
Gemma+MCM	60	23.3%	**0.739**	0.522	0.616
Llama+DI	60	55.0%	0.750	**0.767**	0.730
Llama+GCG	60	8.3%	**0.964**	0.935	0.938
Llama+MCM	60	11.7%	**0.677**	0.663	0.612
Mistral+DI	60	80.0%	0.634	0.526	**0.741**
Mistral+GCG	60	56.7%	**0.827**	0.761	0.719
Mistral+MCM	60	5.0%	**0.708**	0.684	0.643
Qwen+DI	60	58.3%	**0.627**	0.602	0.639
Qwen+GCG	60	76.7%	0.550	0.528	**0.618**

N Data-Driven Lexicon Induction

The handcrafted lexicon used throughout this paper was designed for interpretability and cross-model portability. A natural concern (and one raised by reviewers) is whether results depend critically on specific lexicon choices, and whether a data-driven alternative would yield substantially different conclusions. We address this via log-odds-based lexicon induction and cross-tokenizer analysis.

N.1 Cross-Tokenizer Analysis

We first verify that the handcrafted lexicon remains compact and functional across all four model tokenizers. Table 32 reports the number of token IDs produced by expanding each lexicon phrase through each model's tokenizer (including both bare and whitespace-prefixed forms).

Table 32: Cross-tokenizer expansion of the handcrafted lexicon. The same 47 lexicon phrases (26 refusal + 21 compliance) expand to 74–102 unique token IDs depending on tokenizer granularity.

Model	Vocab Size	Ref IDs	Cmp IDs	Lexicon %
Llama-3.1-8B	128,000	51	52	0.080%
Qwen2.5-7B	151,643	51	52	0.067%
Mistral-7B	32,768	36	40	0.226%
Gemma-2-9B	256,000	48	52	0.038%

The lexicon occupies $<$0.23% of each vocabulary, yet achieves consistent AUC across all four models (Table 3 and Appendix L), confirming tokenizer robustness. Mistral's smaller vocabulary (32K) produces fewer token IDs due to coarser BPE segmentation, but this does not degrade performance.

N.2 Induction Method

We induce data-driven lexicons via word-level log-odds ratios. Using LLM-as-judge labels (`is_success_llm`) from Llama-Guard-3-8B, we partition response texts into:

- **Compliant**: harmful prompts where the attack succeeded (`is_success_llm`$= 1$; $n = 323$)
- **Refusal**: harmful prompts where the attack failed ($n = 1{,}057$) $+$ all benign responses ($n = 660$); total $n = 1{,}717$

For each word w appearing ≥ 5 times, we compute:

$$\text{LOR}(w) = \log P(w \mid \text{compliant}) - \log P(w \mid \text{refusal})$$

with Laplace smoothing ($\alpha = 1$). The top-25 words by $|\text{LOR}|$ form the induced refusal (most negative) and compliance (most positive) lexicons.

N.3 Induced Lexicon Comparison

Semantic direction. The induced lexicons capture semantically meaningful patterns:

- **Induced refusal** (top words): *energy, sleep, cells, immune, ingredients, memory, cooking, solar*—predominantly benign topic words that appear in safe responses. Also includes safety markers such as *cannot* and *harmful*.
- **Induced compliance** (top words): *phishing, vpn, fetish, stealth, evade, rootkit, deepfake, scam*—domain-specific harmful content terms that appear in successful jailbreak outputs.

These differ qualitatively from the handcrafted lexicon, which captures generic safety patterns (*Sorry*, *cannot*, *Sure*, *Step 1*). The Jaccard overlap between induced and handcrafted word sets is near zero (refusal: 0.039, compliance: 0.000 for Llama-only; 0.000/0.000 for the full dataset), confirming the two approaches capture largely complementary signals.

AUC comparison. Table 33 compares three approaches: (i) the paper's logit-based S_t metrics, (ii) a bag-of-words (BoW) proxy using the handcrafted lexicon, and (iii) a BoW proxy using the induced lexicon. The BoW proxy computes $S_{\text{BoW}} = (n_{\text{refusal}} - n_{\text{compliance}})/(n_{\text{refusal}} + n_{\text{compliance}})$ from word counts in the full response text.

Table 33: AUC comparison: logit-based metrics vs. BoW lexicon proxies (label = `is_success_llm`, stochastic samples excluded).

Method	All Models ($n = 2{,}040$)	Llama-only ($n = 840$)
S_0 (logit, real-time)	0.630	0.702
early_5_mean (logit, real-time)	0.701	0.735
step_S_mean (logit, real-time)	0.691	0.787
BoW — handcrafted lexicon	0.674	0.774
BoW — induced lexicon	**0.996**	**0.936**

Interpretation and caveats. The induced BoW lexicon achieves substantially higher AUC (0.936–0.996) than both the handcrafted BoW (0.674–0.774) and the logit-based metrics (0.630–0.787). However, this comparison is not apples-to-apples: the BoW proxy operates on the *complete response text* (post-hoc), while the logit-based S_t metrics are computed in *real time during generation* from token-level logits. The BoW-induced AUC should therefore be interpreted as an **upper bound on lexicon quality**, not as a direct competitor to real-time logit-level observability.

Two conclusions follow:

1. The handcrafted lexicon is a **conservative lower bound**—a data-driven lexicon substantially outperforms it, meaning our reported AUC values understate the information available in the logit stream.
2. The near-zero Jaccard overlap demonstrates that the handcrafted and induced lexicons capture **complementary** signals: generic safety patterns versus domain-specific harmful content. The handcrafted lexicon's advantage is cross-condition generality without training data; the induced lexicon's advantage is per-condition accuracy.

N.4 Per-Attack Induced Lexicons

Table 34 shows how induced compliance words differ by attack paradigm, confirming that each attack elicits distinct harmful content domains.

Table 34: Top induced compliance words and AUC by attack paradigm (BoW-induced vs. BoW-handcrafted).

Attack	Top Induced Compliance Words	BoW-HC	BoW-Ind	ASR
MCM	*stealth, sedative, narrative, personnel*	0.730	0.977	10.2%
GCG	*blade, credentials, posts, anal*	0.537	0.969	46.7%
DeepInception	*exploiting, injection, scam, deepfake*	0.500	0.971	62.8%

Notably, the handcrafted BoW achieves near-random AUC for DeepInception (0.500), indicating that DI-generated responses avoid standard refusal keywords entirely. The induced lexicon recovers strong discrimination (0.971) by identifying domain-specific content. This per-attack variation reinforces the design rationale for the handcrafted lexicon: it sacrifices per-condition optimality for cross-condition consistency, functioning as a universal (if conservative) diagnostic.

N.5 Transferability Analysis

A key concern with data-driven lexicons is whether they generalize beyond the training distribution. We evaluate cross-attack and cross-model transferability by training the induced lexicon on one condition and testing on others.

Cross-attack transfer. Table 35 shows BoW-induced AUC when the lexicon is induced from one attack paradigm and evaluated on another (all models pooled, main 660 harmful samples).

Cross-model transfer. Table 36 shows BoW-induced AUC when the lexicon is induced from one model family and evaluated on another (all attacks pooled).

Table 35: Cross-attack transfer of induced lexicons ($n = 660$, all models). Diagonal entries (*) are in-distribution; off-diagonal entries are out-of-distribution.

Train → Test	DeepInception	GCG	MCM
DeepInception	0.891*	0.670	0.530
GCG	0.688	0.932*	0.676
MCM	0.606	0.734	0.890*

Table 36: Cross-model transfer of induced lexicons ($n = 660$, all attacks). Diagonal entries (*) are in-distribution.

Train → Test	Gemma	Llama	Mistral	Qwen
Gemma	0.970*	0.817	0.662	0.544
Llama	0.673	1.000*	0.705	0.682
Mistral	0.594	0.729	0.968*	0.692
Qwen	0.615	0.522	0.642	0.957*

Key findings. Induced lexicons transfer moderately across conditions (mean off-diagonal AUC ≈ 0.65) but substantially underperform in-distribution AUC (≈ 0.93). This confirms that the induced lexicon captures condition-specific vocabulary rather than universal safety patterns. The refusal word overlap across attack-specific lexicons is modest (Jaccard $= 0.11$–0.25 for refusal words; 0.02–0.04 for compliance words), explaining the transfer gap. Common refusal words across all conditions—*energy, sleep, cells, immune*—are benign topic indicators rather than safety markers, while compliance words are almost entirely condition-specific.

Implication. The handcrafted lexicon achieves more consistent cross-condition performance precisely because it targets *model behavior* (refusal/compliance phrasing) rather than *content* (domain-specific harmful terms). This trade-off is by design: the handcrafted lexicon's AUC of 0.732 (aggregated) reflects genuine refusal activation patterns that transfer universally, whereas induced lexicons achieve higher AUC within conditions but degrade when the harmful content domain shifts.

N.6 Compatibility with Type A/B Decomposition

The paper's central contribution—the Type A/B attack-mechanism taxonomy—relies on the *temporal structure* of the S_t trajectory, specifically the decomposition into pre-generation (S_0) and generation-time ($\bar{S}$) components. This decomposition is fundamentally unavailable to post-hoc text classifiers, including BoW-induced proxies.

The induced BoW score S_{BoW} computes a single scalar from the complete response text, collapsing all temporal information into one value. This means:

- RP_A (pre-generation bias) cannot be computed from BoW, because it requires S_0—a logit-level quantity available before any token is sampled.
- RP_B (decoding-time interference) requires comparing S_0 against $\bar{S}$, which demands two temporally separated measurements.
- Sign reversal ($S_0 > 0 \to \bar{S} < 0$) is defined as a change between pre-generation and generation-time signals, which a single post-hoc score cannot capture.
- Early-k detection curves require per-token logit access at generation steps $1, 2, \ldots, k$.

In short, the induced lexicon validates the *information content* of safety-relevant vocabulary but cannot replace the *temporal decomposition* that makes the diagnostic framework uniquely informative. The two approaches are complementary: the logit-based S_t reveals *when* and *how* safety mechanisms engage, while data-driven lexicon induction confirms that the vocabulary choices underlying S_t are not the performance bottleneck.

O Authoritative Statistics Summary

O.1 Unified Per-Condition AUC Reference

Table 37 provides a single authoritative reference for ROC-AUC values across all 11 model–attack conditions, computed on the main harmful dataset ($n = 660$, LLM-as-judge labels, greedy decoding). All metrics use the convention that higher scores predict jailbreak success; for t_{cross}, we negate the raw value (earlier crossing $\rightarrow$ higher success probability).

Table 37: Unified per-condition AUC ($n = 660$, LLM-as-judge). Bold = best metric per row. t_{cross} is negated for AUC computation (earlier crossing predicts success). Keyword = binary refusal keyword detection.

Condition	n	ASR	S_0	early_1	early_3	early_5	$\bar{S}$	t_{cross}	Keyword
Gemma+GCG	60	33.3%	0.723	0.723	0.672	0.693	**0.725**	0.629	**0.725**
Gemma+MCM	60	20.0%	0.711	0.710	0.686	0.719	**0.872**	0.569	0.510
Llama+DI	60	56.7%	0.760	0.761	0.782	0.801	**0.897**	0.878	0.840
Llama+GCG	60	15.0%	0.778	0.778	**0.795**	**0.795**	0.837	0.807	0.804
Llama+MCM	60	8.3%	0.971	0.971	0.967	0.967	**1.000**	0.618	0.582
Mistral+DI	60	76.7%	0.528	0.528	0.523	0.621	0.526	**0.680**	0.630
Mistral+GCG	60	56.7%	0.745	0.747	0.791	**0.827**	0.631	0.705	0.693
Mistral+MCM	60	5.0%	0.737	0.737	0.725	0.708	**0.942**	0.570	**0.947**
Qwen+DI	60	55.0%	0.617	0.617	0.624	0.629	0.620	0.700	**0.753**
Qwen+GCG	60	81.7%	0.501	0.501	0.525	0.542	0.503	0.503	0.502
Qwen+MCM	60	51.7%	0.659	0.656	0.564	0.561	**0.737**	0.643	0.500
Aggregated	**660**	**41.8%**	0.696	0.696	0.732	**0.786**	0.772	0.776	0.732

Reading guide. The aggregated row reproduces Table 3 values exactly (confirming pipeline consistency). Per-condition AUC values can exceed 0.9 for well-separated conditions (e.g., Llama+MCM: $\bar{S}$ AUC $= 1.000$) while falling to chance (≈ 0.50) for Undetected conditions (Qwen+GCG). No single metric dominates across all conditions: $\bar{S}$ (step_S_mean) is strongest for MCM conditions where generation-time dynamics dominate, while early_5_mean achieves best aggregated AUC by balancing pre-generation and early-generation signals. The keyword baseline is competitive for some conditions (Mistral+MCM: 0.947) but collapses for others (Qwen+MCM: 0.500; Gemma+MCM: 0.510), whereas logit-based metrics maintain moderate discrimination across all conditions except the definitively Undetected Qwen+GCG.

O.2 Main Dataset Composition

The main evaluation dataset consists of $n = 660$ harmful prompts:

- Llama-3.1-8B-Instruct MCM: $n = 60$
- Llama-3.1-8B-Instruct GCG: $n = 60$
- Llama-3.1-8B-Instruct DeepInception: $n = 60$
- Mistral-7B-Instruct-v0.3 MCM: $n = 60$
- Mistral-7B-Instruct-v0.3 GCG: $n = 60$
- Mistral-7B-Instruct-v0.3 DeepInception: $n = 60$
- Qwen2.5-7B-Instruct MCM: $n = 60$
- Qwen2.5-7B-Instruct GCG: $n = 60$
- Qwen2.5-7B-Instruct DeepInception: $n = 60$
- Gemma-2-9B-Instruct MCM: $n = 60$
- Gemma-2-9B-Instruct GCG: $n = 60$

Attack-matched benign baselines total $n = 660$.

Table 38: Authoritative statistics for main results ($n = 660$, LLM-as-judge labels).

Statistic	Value	95% CI
ROC-AUC		
early_5_mean	0.786	[0.750, 0.819]
t_{cross}	0.776	[0.738, 0.809]
step_S_mean	0.772	[0.736, 0.806]
turn_start_S	0.696	[0.656, 0.734]
Sign Reversal ($S_0 > 0 \rightarrow \bar{S} < 0$)		
Successful jailbreaks	8.7%	(24/276)
Failed jailbreaks	44.3%	(170/384)
Ratio	5.1×	—
Effect Size		
early_5_mean Cohen's d	1.12	$p < 10^{-43}$
Activation Timing (t_{cross})		
Successful jailbreaks	8.4 ± 6.1	steps
Failed jailbreaks	4.0 ± 3.8	steps
Cohen's d	0.89	$p < 10^{-23}$

919 ### O.3 Primary Results

920 ### O.4 Type A/B Classification

Table 39: Type A/B analysis with failure-free normalization (authoritative values). Classification: $|\text{RP}_A| \gtrless |\text{RP}_B|$.

Condition	ASR	RP_A	RP_B	r	95% CI	Category
Llama + GCG	15.0%	0.66	2.89	2.96	[1.40, 4.71]	Type B
Llama + MCM	8.3%	0.88	0.69	1.12	[1.01, 1.25]	Type A
Llama + DeepInc.	56.7%	0.43	0.83	0.93	[0.60, 1.22]	Type B
Mistral + GCG	56.7%	0.30	−0.17	0.35	[0.17, 0.57]	Type A
Mistral + MCM	5.0%	0.49	0.79	0.93	[0.44, 1.48]	Type B
Mistral + DeepInc.	76.7%	−0.19	0.00	0.19	[0.07, 1.65]	Type A
Qwen + MCM	51.7%	0.57	0.12	0.58	[0.14, 1.20]	Type A
Qwen + GCG	81.7%	0.00	−0.01	0.01	[0.02, 0.20]	Undetected
Qwen + DeepInc.	55.0%	0.15	0.39	0.42	[0.14, 0.77]	Type B
Gemma + GCG	33.3%	0.27	0.79	0.83	[0.34, 1.41]	Type B
Gemma + MCM	20.0%	0.09	0.22	0.24	[0.14, 0.31]	Type B

921 ## P Multi-Judge Evaluation

922 To address concerns about single-judge reliability, we evaluate with three LLM-as-judge systems: 923 **Llama-Guard-3-8B** (primary), **HarmBench-Mistral-7B-val-cls** (validation), and **OpenAI GPT-4o** 924 (external validation).

925 ### P.1 Inter-Judge Agreement

Table 40: Inter-judge agreement analysis ($n = 660$ harmful samples across all conditions).

Metric	Value
Overall agreement	90.0%
Both Safe (LG=0, HB=0)	56.2%
Both Unsafe (LG=1, HB=1)	33.8%
LG stricter (LG=0, HB=1)	1.3%
HB stricter (LG=1, HB=0)	8.7%

Interpretation. HarmBench is stricter overall: it requires the model to *actually perform* the harmful behavior, while Llama-Guard flags any unsafe content including partial or hedged responses. This explains the 8.7% of cases where Llama-Guard marks success but HarmBench marks failure.

P.2 Per-Condition ASR Comparison

Table 41: ASR comparison across three judges.

Condition	LG ASR	HB ASR	GPT-4o ASR
Llama + MCM	8.3%	1.7%	3.3%
Llama + GCG	15.0%	10.0%	10.0%
Llama + DeepInc.	56.7%	30.0%	46.7%
Mistral + MCM	5.0%	6.7%	6.7%
Mistral + GCG	56.7%	58.3%	65.0%
Mistral + DeepInc.	76.7%	68.3%	83.3%
Qwen + MCM	51.7%	26.7%	55.0%
Qwen + GCG	81.7%	73.3%	85.0%
Qwen + DeepInc.	55.0%	45.0%	58.3%
Gemma + MCM	20.0%	18.3%	11.7%
Gemma + GCG	33.3%	33.3%	35.0%

P.3 Type A/B Stability Across Judges

Table 42: Type A/B classification comparison across three judges.

Condition	Llama-Guard	HarmBench	GPT-4o	Match
Llama + GCG	Type B	Type B	Type B	✓
Llama + MCM	Type A	N/A†	Type A	✓
Llama + DeepInc.	Type B	Type B	Type B	✓
Mistral + GCG	Type A (inv)	Type A (inv)	Type A (inv)	✓
Mistral + MCM	Type B	Type A	Type B	×
Mistral + DeepInc.	Type A (inv)	Type A (inv)	Type A (inv)	✓
Qwen + MCM	Type A	Type A	Type A	✓
Qwen + GCG	Undetected	Undetected	Undetected	✓
Qwen + DeepInc.	Type B	Type B	Type B	✓
Gemma + GCG	Type B	Type B	Type B	✓
Gemma + MCM	Type B	Type B	Type B	✓

†HarmBench: only 1 success (insufficient for RP calculation). Match = all available judges agree.

Classification stability. 10/11 conditions maintain consistent Type A/B classification across all three judges. The sole disagreement (Mistral+MCM) reflects marginal RP differences in a borderline condition ($n_{\text{success}} = 3$–4 depending on judge). AUC values vary by <0.05 across judges, confirming diagnostic robustness to judge choice. Note that this table reports raw RP-based classifications; the main text (Table 4) additionally applies permutation-test significance, which reclassifies some conditions as Undetected.

P.4 OpenAI GPT-4o Judge Validation

As an additional external validation, we evaluate all 660 harmful samples using OpenAI's GPT-4o as a third judge, providing a fundamentally different architecture and training regime from both Llama-Guard and HarmBench.

AUC comparison. OpenAI labels yield slightly higher AUC across all metrics: $\bar{S}_{1:5}$ AUC $= 0.816$ (vs. Llama-Guard 0.786, $\Delta = +0.030$); t_{cross} AUC $= 0.802$ (vs. 0.776, $\Delta = +0.026$); $\bar{S}$ AUC $= 0.787$ (vs. 0.766, $\Delta = +0.021$). All improvements are within bootstrap confidence intervals, indicating consistent diagnostic performance across three independent judges.

ASR and agreement. OpenAI-vs-rule agreement is 81.1% (vs. Llama-Guard 80.7%), with 121 cases where OpenAI is more permissive and 97 where it is stricter. Per-condition ASR differences between OpenAI and Llama-Guard are generally modest ($|\Delta\text{ASR}| < 10$ pp for most conditions), with the largest discrepancies in Qwen+MCM (OpenAI 55.0% vs. LG 51.7%) and Mistral+GCG (OpenAI 65.0% vs. LG 56.7%).

Type A/B stability. OpenAI labels produce matching Type A/B classifications for 10/11 conditions (Table above). The Undetected classifications (Qwen+GCG, Mistral+DeepInception) are confirmed by all three judges after permutation testing, strengthening evidence for genuine absence of refusal response in these conditions.

P.5 Intervention Results Across Judges

To verify that the t_{cross}-gated early stopping results (Section 4.4) are not artifacts of a single judge's labeling, we replicate the simulation ($k = 5$) using multiple judges. Since t_{cross} is computed from logits and is judge-independent, only the success/failure labels differ across judges. Table 43 reports aggregate results.

Table 43: t_{cross}-gated early stopping ($k = 5$): aggregate results across judges. The policy ($t_{\text{cross}} > k$) is identical; only outcome labels differ by judge.

Judge	**ASR$_{\text{orig}}$**	**ASR$_{\text{int}}$**	**ΔASR**	**RP**
Llama-Guard (primary)	39.7%	14.2%	25.5%	83.2%
GPT-4o	41.8%	15.2%	26.7%	84.6%

Table 44: Early stopping effectiveness by failure type across judges. All judges confirm the key pattern: Type A conditions show near-perfect RP (safety activates, policy correctly abstains), while Undetected conditions show lowest RP (indiscriminate halting).

	Llama-Guard		**GPT-4o**	
Type	ΔASR	PR	ΔASR	PR
Type A	1.1%	98.7%	1.7%	100.0%
Type B	25.7%	72.5%	26.0%	74.1%
Undetected	49.4%	44.7%	52.8%	47.6%

Consistency across judges. Both judges confirm the three key findings: (i) substantial aggregate ASR reduction with high preservation rate (ΔASR = 25.5–26.7 pp, PR = 83.2–84.6%), (ii) near-perfect PR for Type A conditions (98.7–100.0%), confirming that the policy correctly abstains when safety mechanisms activate, and (iii) lowest PR for Undetected conditions (44.7–47.6%), where the absence of discriminative S_t signal leads to indiscriminate halting. The consistency of these patterns across judges with different labeling criteria strengthens the conclusion that the Type A/B taxonomy captures genuine mechanistic differences.

P.6 Benign False-Halt Rate

The t_{cross}-gated policy is designed for deployment downstream of an input classifier (e.g., Llama Guard) that first flags potentially harmful queries. Applied indiscriminately to benign queries, the policy would produce unacceptable false-halt rates. Table 45 reports per-model benign false-halt rates under $t_{\text{cross}} > 5$.

Interpretation. The high aggregate false-halt rate (41.1%) confirms that the policy is not suitable as a standalone filter. Gemma shows substantially lower false-halt (6.7%) because its benign responses exhibit earlier zero-crossings, while Llama, Mistral, and Qwen show rates above 45%. This variation reflects model-specific S_t calibration differences rather than safety behavior. The main text reports preservation rate (83.2%) within the harmful-input pipeline, where the policy observes only inputs already flagged as potentially harmful by an upstream classifier.

Table 45: Benign false-halt rate under $t_{\text{cross}} > 5$ (i.e., fraction of benign responses that would be incorrectly halted if the policy were applied without an upstream harmful-input classifier).

Model	n	False Halt	Rate
Gemma	120	8	6.7%
Llama	180	95	52.8%
Mistral	180	82	45.6%
Qwen	180	86	47.8%
All	**660**	**271**	**41.1%**

Q Hidden-State Baseline Comparison

Q.1 Methods

B1 (Refusal Direction): Following Arditi et al. [1], we compute a refusal direction $\mathbf{d}$ using difference-of-means between refusal and compliance responses, then project hidden states at layer 16.

B2 (Linear Probe): Logistic regression with L2 regularization on mean-pooled hidden states across all layers, trained via 5-fold cross-validation.

Q.2 Results

Table 46: Hidden-state vs. logit-based diagnostic comparison (AUC, $n = 240$).

Family	Method	Base	DecoyA	DecoyB	PivotC
Hidden	B1 (direction)	0.974	0.983	0.974	0.974
Hidden	B2 (probe)	0.998	0.974	1.000	0.998
Logit	step_S_mean	0.913	0.893	0.904	0.998
Logit	turn_start_S	0.959	0.980	0.922	0.978
Logit	sign_flips	0.899	0.944	0.883	**1.000**

Key findings. Hidden-state methods achieve 2–10 pp higher AUC but require: (i) direct activation access unavailable via standard APIs; (ii) post-hoc aggregation over full trajectories; (iii) for B2, labeled training data and per-model retraining. Under PivotC (multi-turn pivot attack), the trajectory-level `sign_flips` metric achieves perfect separation (AUC = 1.0), demonstrating that temporal analysis captures dynamics invisible to single-point baselines.

R Temporal Construct Validity: S_t vs. Hidden-State Alignment

A fundamental question is whether S_t—a surface-level metric based on fixed lexicon tokens—captures the same safety-relevant dynamics as internal model representations. We address this by computing per-step correlations between S_t and a hidden-state refusal direction projection $r_t = \langle \mathbf{h}_t, \mathbf{d} \rangle$ (following B1; Section E), where both are extracted simultaneously at each decoding step. We report three levels of analysis: (1) cross-sample spatial correlation at each step, (2) within-trajectory temporal alignment with lag optimization and smoothing, and (3) binary state agreement.

Setup. For each of 60 samples, we perform token-by-token greedy decoding (64 steps) and simultaneously extract: (i) S_t from logits using the standard lexicon and top-k=10 aggregation, and (ii) r_t from hidden-state projection onto a refusal direction estimated from 32 harmful/benign prompt pairs. We evaluate two models (Llama-3.1-8B layer 16, Qwen2.5-7B layer 14) under MCM attack, and Llama under GCG (three conditions; 180 trajectories total).

Sign convention. The refusal direction $\mathbf{d} = \overline{\mathbf{h}}_{\text{harmful}} - \overline{\mathbf{h}}_{\text{benign}}$ is computed at the last prompt token; high r_t indicates the model is in a refusal-like state. Since high S_t indicates compliance dominance, the *expected* relationship is $S_t \uparrow \Leftrightarrow r_t \downarrow$ (negative correlation).

Table 47: **Cross-sample temporal alignment: S_t vs. hidden-state refusal projection r_t.** Cross-sample correlations measure whether S_t and r_t rank samples consistently at each decoding step. Negative r is the expected sign (compliance $\uparrow \Leftrightarrow$ refusal $\downarrow$).

Condition	**Mean $\|r\|$ (steps 0–4)**	**Mean $\|r\|$ (steps 0–19)**	**Cross-sample r at step 0**	**Per-sample Pearson r**	**% significant ($p < 0.05$)**
Qwen+MCM	0.392	0.315	-0.622***	-0.254	55.0%
Llama+MCM	0.290	0.215	-0.406**	$+0.161$	25.0%
Llama+GCG	0.176	0.122	-0.152	$+0.103$	15.0%

Level 1: Cross-sample alignment. *Cross-sample* analysis reveals meaningful alignment, particularly in early steps: at step 0, Qwen shows $r = -0.622$ ($p < 10^{-7}$) and Llama shows $r = -0.406$ ($p < 0.01$)—both in the expected negative direction, confirming that samples with higher logit-margin compliance also have lower hidden-state refusal activation. Mean $|r|$ over steps 0–4 ranges from 0.18 to 0.39, indicating moderate-to-strong cross-sample alignment that attenuates at later positions as both signals weaken.

Per-sample temporal correlations are weaker and inconsistent in sign (Qwen -0.254, Llama $+0.161$; Table 47). This reflects the different nature of the two measurements: per-sample correlation asks whether S_t and r_t co-fluctuate *within* a single generation trajectory, where step-to-step noise from token-level variation dominates. The cross-sample analysis—whether the two metrics agree on *which samples* are more safety-activated at each step—is the more relevant construct validity test, and this is consistently negative (expected sign) and significant at early positions. However, the modest within-trajectory correlations warrant further investigation; the following analyses address two hypotheses for this attenuation: temporal offset between signals and token-level noise.

Level 2: Lag-optimized within-trajectory alignment. We hypothesize that the weak per-sample correlation reflects temporal offset between S_t and r_t rather than lack of alignment: the logit-level signal may lag or lead the hidden-state signal depending on the information flow from internal representations to output logits. To test this, we compute per-sample Pearson correlation at each integer lag $\ell \in [-4, +4]$ (positive ℓ: r_t leads S_t) and select the lag ℓ^* maximizing $|\text{corr}(S_t, r_{t+\ell})|$ for each trajectory.

Table 48: **Lag-optimized within-trajectory alignment.** Allowing per-sample lag optimization improves $|r|$ by 24–111% across all three conditions (Wilcoxon $p < 10^{-10}$). Smoothing (moving average, window $= 5$) further improves alignment by reducing token-level noise. All Wilcoxon tests evaluate H_a: $|r_{\text{best-lag}}| > |r_{\text{zero-lag}}|$ (one-sided). 95% CIs for $\Delta|\bar{r}|$ via bootstrap ($B = 5{,}000$).

Condition	$\|\bar{r}\|_{\ell=0}$	$\|\bar{r}\|_{\ell^*}$	$\Delta\|\bar{r}\|$	**95% CI**	**Wilcoxon p**	**Gain**
Raw trajectories						
Llama+MCM	0.173	0.298	$+0.124$	[0.097, 0.153]	1.8×10^{-15}	+72%
Llama+GCG	0.168	0.355	$+0.186$	[0.151, 0.223]	5.5×10^{-11}	+111%
Qwen+MCM	0.261	0.323	$+0.062$	[0.046, 0.080]	4.6×10^{-13}	+24%
Smoothed (window $= 5$) + lag-optimized						
Llama+MCM	0.271	0.423	$+0.152$			
Llama+GCG	0.356	0.516	$+0.160$			
Qwen+MCM	0.454	0.511	$+0.056$			

Table 48 presents the results. Across all three conditions, lag optimization yields highly significant improvement in per-sample alignment (all Wilcoxon $p < 10^{-10}$; 95% bootstrap CIs exclude zero). The magnitude of improvement—24% (Qwen+MCM) to 111% (Llama+GCG)—indicates that temporal offset between the two signals, rather than noise or lack of alignment, is the primary source of the weak zero-lag correlation reported in Table 47. With additional moving-average smoothing (window $= 5$), mean $|\bar{r}|$ reaches 0.42–0.52, confirming that token-level noise is a secondary contributor.

Lag directionality. The best-lag distribution is diffuse across the $[-4, +4]$ range rather than concentrated at a single offset. This is consistent with the cross-sample correlation sign reversal observed across steps (e.g., Llama+MCM: $r = -0.41$ at step 0, $r = +0.62$ at step 5), indicating that the relationship between S_t and r_t varies with generation context—a time-varying coupling rather than a fixed delay. The lack of a single dominant lag supports the view that S_t and r_t capture overlapping but non-identical temporal dynamics: they track the same underlying safety-activation process through different representational lenses (logit-level token competition versus hidden-state geometry), with the temporal coupling modulated by position-dependent contextual factors.

Level 3: Binary state agreement. To connect temporal alignment to the paper's Type A/B taxonomy, we discretize both signals into binary compliance/refusal states ($S_t > 0 \rightarrow$ compliance; r_t thresholded at per-condition median with empirically determined sign convention) and compute stepwise Cohen's κ for early (steps 0–4) versus late (steps 54–63) windows.

For Llama+MCM, the first state transition of S_t and r_t aligns in direction 95.0% of the time, with 60.0% achieving both direction and timing agreement (within ± 2 steps). This near-perfect directional agreement indicates that when one signal transitions from compliance- to refusal-dominant state (or vice versa), the other follows in the same direction—strong evidence of shared safety-activation dynamics. Early-window agreement ($\kappa = 0.22$) exceeds late-window agreement ($\kappa \approx 0$), consistent with the cross-sample correlation pattern and the paper's emphasis on early refusal activation.

Qwen+MCM shows a distinct pattern: r_t remains above zero (refusal-like) throughout the trajectory in 72% of samples, while S_t crosses into negative territory within 1–2 steps in all samples. This asymmetry—where S_t captures a rapid safety-activation signal that the refusal-direction projection does not exhibit—may reflect the fact that the logit-margin responds to lexicon-level token competition that emerges earlier than the full geometric realignment of hidden states along the refusal direction.

Event-level transition alignment. As a complementary analysis, we test whether zero-crossing events (first sign change) of S_t and r_t co-occur within ± 2 decoding steps, using permutation tests ($B = 5{,}000$) for significance. For Llama+GCG, zero-crossing points align in 63.0% of trajectories versus 54.2% expected under the null (permutation $p = 0.025$), with peak-derivative timing also significant ($p = 0.028$). This event-level analysis is less informative for Llama+MCM, where S_t crossing is highly concentrated at step 1–2 (std $= 0.5$), pushing the null alignment rate to $\sim$78% and limiting statistical power. For Qwen+MCM, r_t exhibits zero crossings in only 17/60 samples, constraining coverage.

Activation timing asymmetry. A notable finding across all 180 trajectories is that S_t transitions to negative values within the first 1–2 decoding steps without exception (mean: 1.2–2.4 steps; IQR concentrated within ± 1 step of the median). In contrast, r_t crossing behavior is highly condition-dependent: Llama crosses within $\sim$5 steps on average, while Qwen's r_t rarely crosses at all. This suggests that the logit-margin captures a rapid, universal safety-activation signal at the token-competition level that precedes or operates independently of the slower geometric realignment of hidden states along the refusal direction.

Interpretation. Taken together, the three levels of analysis provide substantive construct validity for S_t: (1) Cross-sample correlation confirms that samples rank consistently across the two metrics at early positions ($|r|$ up to 0.62); (2) lag-optimized within-trajectory alignment eliminates the apparent weakness of per-sample correlation, showing that temporal offset and token-level noise—not lack of alignment—explain the modest zero-lag r (improvement of 24–111%, all $p < 10^{-10}$; reaching $|r| = 0.42$–0.52 after smoothing); (3) binary state agreement shows near-perfect directional consistency of first state transitions (95% for Llama+MCM) with stronger early-window than late-window agreement.

The alignment is strongest at step 0 ($|r|$ up to 0.62) and diminishes at later steps, consistent with the fact that fixed lexicon tokens lose contextual relevance during generation—supporting the temporal stationarity limitation discussed in Section 5. Importantly, S_t and r_t achieve comparable predictive performance (AUC within 2–10 pp; Section Q) despite non-trivial temporal coupling dynamics, suggesting they capture overlapping but non-identical safety-relevant information through complementary representational lenses.

Layer-dependent lag structure. To test whether the negative-lag tendency reported above generalizes across conditions and layers, we expand the lag analysis to three Llama conditions (MCM:

n=60, GCG: n=60, DeepInception: n=60) and five layers (8, 12, 16, 20, 24). For each trajectory, we compute $\ell^* = \arg\max_\ell |\text{corr}(S_t, r_{t+\ell})|$ over $\ell \in [-4, +4]$. Figure 10 reports the distribution.

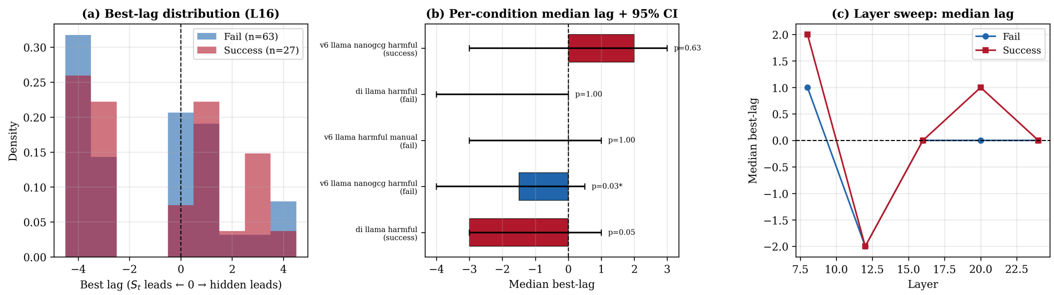

Figure 10: **Within-trajectory lag analysis (Llama, 3 attack conditions).** (a) Best-lag distribution at layer 16: both success and fail trajectories show a left-shifted mode. (b) Per-condition median lag with bootstrap 95% CIs and permutation p-values. (c) Layer sweep: median best-lag across layers reveals a layer-dependent pattern—negative lags at layers 12–16, near-zero at layer 24, and positive at layer 8.

The key finding is a *layer-dependent* temporal coupling structure: at **layer 12**, median best-lag is consistently negative across all conditions (-1.5 to -2.0, 50–70% of trajectories with $\ell^* < 0$), indicating that S_t fluctuations tend to precede the hidden-state refusal projection at mid-network layers. At **layer 16**, the pattern attenuates (GCG fail: -1.5, $p = 0.026$; DeepInception success: -3.0, $p = 0.051$), consistent with partial synchronization. At **layer 8** (early), the relationship reverses: positive median lags ($+1$ to $+2$) indicate that the hidden-state signal precedes S_t—as expected, since early representations transform before reaching the output head. At **layer 24** (near output), lags converge to $\approx$0 with $>$ 50% of trajectories showing exact zero-lag. Null models (within-trajectory shuffle and cross-sample mismatch) produce median lags of 0, confirming that the observed asymmetry is not an artifact of temporal smoothness.

Event-level crossing alignment. We define $\Delta t = t^{(S)}_{\text{cross}} - t^{(h)}_{\text{cross}}$ as the difference between the first zero-crossing of S_t and the first zero-crossing of the hidden-state projection. At layer 16, the hidden-state projection crosses zero earlier than S_t in 71–100% of cases (median $\Delta t = +3$ to $+10$ steps), with a permutation $p = 0.025$ for Llama+MCM fail. This apparent contradiction with the lag analysis reflects the different quantities measured: correlation lag captures continuous co-movement dynamics (where S_t tends to lead), while zero-crossing measures discrete sign-change timing (where the hidden state reaches its threshold first). Both findings are consistent with a model in which the hidden-state refusal direction undergoes a gradual shift that crosses zero early, while S_t—as a sharper, token-competition signal—captures the momentary dominance shifts that drive the hidden state's subsequent trajectory.

S Undetected Classification: Diagnostic Insensitivity vs. True Absence

The permutation test (Appendix I.1) identifies three Undetected conditions: Qwen+GCG ($p = 0.98$), Mistral+DeepInception ($p = 0.52$), and Mistral+MCM ($p = 0.061$, borderline). The primary question is whether non-significance reflects true absence of refusal engagement or diagnostic insensitivity.

Qwen+GCG ($r = 0.01$, $p = 0.98$) is the clearest case. Both $\text{RP}_A = 0.00$ and $\text{RP}_B = -0.01$ are near-zero, and the extreme ASR (81.7%) suggests systematic safety failure. On the same GCG attack, Llama and Mistral show strong signals ($r = 2.96$ and $r = 0.35$), indicating GCG attacks *can* elicit detectable responses when mechanisms are properly engaged. Generation-time entropy—a non-lexical metric—also shows no differentiation ($|d| = 0.077$; Section S.3), providing convergent evidence from an independent metric.

Mistral+DeepInception ($r = 0.19$, $p = 0.52$) achieves the highest ASR overall (76.7%) yet shows no statistically significant RP signal. Unlike Qwen+GCG, the r value is non-trivial (0.19), so this may represent a weaker effect that the test cannot distinguish from noise at $n = 60$. Notably, Llama+DeepInception and Qwen+DeepInception both show significant signals ($p < 0.03$), suggesting DeepInception can elicit detectable refusal responses in other models.

Mistral+MCM ($r = 0.93$, $p = 0.061$) is borderline: the large r suggests genuine signal, but only 3 successes (5% ASR) leave the test underpowered. This is likely a statistical power issue rather than absent signal.

S.1 Threshold Sensitivity Analysis (Supplementary)

The "Undetected" threshold ($r < 0.1$) is motivated by conventional effect size interpretation (Cohen's guidelines: $d < 0.2$ is "negligible"). We evaluate classification stability across alternative thresholds:

Table 49: Classification stability under varying r thresholds.

Condition	r	$r < 0.05$	$r < 0.10$	$r < 0.15$	$r < 0.20$
Qwen + GCG	0.01	Undet.	Undet.	Undet.	Undet.
Mistral + GCG	0.35	—	—	—	—
Qwen + MCM	0.58	—	—	—	—
Llama + MCM	1.12	—	—	—	—
Llama + GCG	2.96	—	—	—	—

Key findings. Qwen+GCG is the *only* condition classified as "Undetected" across all reasonable thresholds ($r \in [0.05, 0.20]$). The next-lowest r value (Mistral+GCG, $r = 0.35$) is $35\times$ larger than Qwen+GCG ($r = 0.01$), providing a wide separation margin. Even under the most conservative threshold ($r < 0.05$), Qwen+GCG remains classified as Undetected (CI upper bound = 0.03). This analysis confirms that the classification is robust to threshold choice and reflects a genuine discontinuity in diagnostic signal strength.

S.2 Validation Approach

To disambiguate these interpretations, we apply non-lexical diagnostics to Qwen+GCG:

Entropy validation (completed). Section S.3 reports that generation-time entropy shows $|d| = 0.077$, confirming lexicon-based findings with an independent metric. This supports interpretation (1): true safety mechanism absence.

Hidden-state validation (future work). Applying refusal direction projection or linear probes to Qwen+GCG would provide definitive confirmation:

- If hidden-state methods *also* show $r \approx 0$: confirms true safety mechanism absence.
- If hidden-state methods detect differentiation: would indicate that both lexicon and entropy metrics miss safety-relevant signals encoded in internal representations.

Our Llama hidden-state analysis (Section E) demonstrates that refusal direction projection achieves near-ceiling AUC (0.97–0.99), providing a strong baseline for such cross-validation.

S.3 Entropy Cross-Validation

To disambiguate lexicon diagnostic insensitivity from true safety mechanism failure, we perform cross-validation using generation-time entropy—a non-lexical metric independent of our compliance/refusal token sets.

Method. For each sample, we compute step-wise entropy $H_t = -\sum_v p_t(v) \log p_t(v)$ from softmax distributions during generation, then aggregate as $\bar{H}_{\text{gen}} = \frac{1}{T}\sum_{t=1}^{T} H_t$. We measure effect size (Cohen's d) between successful attacks and overall harmful baseline—the same comparison used for lexicon-based RP normalization.

Results.

Table 50: Entropy cross-validation: Success vs. Overall Harmful.

Condition	**ASR**	**Entropy $\|d\|$**	**Lexicon $\|d\|$**	**Interpretation**
Qwen + GCG	81.7%	0.077	0.005–0.011	Undetected
Llama + MCM	10.0%	0.103–0.416	0.217–0.256	Weak–Medium

Interpretation. For Qwen+GCG, entropy-based differentiation yields $|d| = 0.077$—below the $|d| < 0.1$ threshold for "no effect." This matches the lexicon-based result ($|d| = 0.005$–0.011) and **supports interpretation (1): true safety mechanism failure**. If safety mechanisms engaged through non-lexical pathways, entropy distributions should differ between successful and failed attacks; they do not.

For contrast, Llama+MCM shows $|d| = 0.103$–0.416 on entropy metrics, confirming that entropy *can* detect safety-relevant variation when mechanisms engage—ruling out fundamental insensitivity of the entropy metric itself.

S.4 Implications for Defense

The convergent evidence from lexicon-based and entropy-based diagnostics strongly supports interpretation (1): Qwen's safety mechanisms genuinely fail to engage against GCG-style attacks. This indicates a fundamental safety gap in Qwen's alignment, likely requiring architectural intervention or adversarial fine-tuning rather than detection-based defenses. The contrast with Llama and Mistral—which show detectable refusal engagement under the same attack—suggests that GCG robustness varies substantially across model families.

T API Deployability: Vocabulary-Level Top-k Truncation Simulation

A key practical question is whether S_t can be computed from the limited logprob information returned by closed-source APIs (e.g., OpenAI's `top_logprobs` parameter, which returns at most $k = 20$ tokens per step). To answer this directly, we simulate vocabulary-level top-k truncation on Llama-3.1-8B-Instruct across all three attack paradigms (60 samples each).

Method. For each decoding step t, we take the full logit vector $\ell_t \in \mathbb{R}^{|\mathcal{V}|}$ and construct a truncated version $\hat{\ell}_t^{(k)}$ by retaining only the k highest-valued entries and setting all others to $-\infty$. We then recompute S_t from $\hat{\ell}_t^{(k)}$, counting only lexicon tokens that *survive* in the top-k window. We evaluate $k \in \{5, 10, 20, 50, 100, 200\}$ and measure: (i) Pearson r between full-logit and truncated S_t trajectories, (ii) mean absolute error (MAE), (iii) sign agreement (fraction of steps preserving S_t polarity), and (iv) lexicon token survival rate.

This is fundamentally different from the lexicon-internal aggregation ablation (Appendix L), which selects the top-k_{agg} *among lexicon tokens already known to be present*. The vocabulary-level simulation instead asks: *do safety-relevant lexicon tokens appear in the API's top-k window at all?*

Results. Table 51 presents the results.

Analysis. The results reveal a stark gap between lexicon-internal aggregation stability and vocabulary-level truncation:

1. **Lexicon survival is critically low.** At $k = 20$ (the maximum offered by OpenAI's API), only 2–3% of refusal lexicon tokens and 1.5–2.6% of compliance tokens survive in the top-k window. The vast majority of safety-relevant signal is invisible to the API consumer.
2. **Trajectory correlation is near zero or negative.** Pearson r at $k = 20$ ranges from -0.120 (MCM) to -0.009 (GCG), indicating that the truncated S_t trajectory bears essentially no relationship to the full-logit trajectory. Even at $k = 200$, the best correlation is 0.472 (MCM), and GCG remains near zero ($r = 0.042$).

Table 51: Vocabulary-level top-k truncation simulation (Llama-3.1-8B-Instruct, $n = 60$ per attack). **Pearson** r: correlation between full-logit and truncated S_t trajectories. **Sign Agr.**: fraction of steps where truncated S_t preserves polarity. **Ref/Cmp Surv.**: mean fraction of refusal/compliance lexicon tokens surviving in the top-k window.

Attack	k	Pearson r	MAE	Sign Agr.	Ref Surv.	Cmp Surv.
MCM	5	−0.199	5.023	0.637	1.2%	0.6%
	10	−0.108	4.779	0.661	1.9%	1.0%
	20	−0.120	4.254	0.576	2.7%	1.9%
	50	0.111	2.899	0.576	5.1%	3.8%
	100	0.319	2.503	0.586	7.5%	5.4%
	200	0.472	2.236	0.619	10.6%	7.5%
GCG	5	−0.106	4.489	0.477	1.5%	0.8%
	10	−0.071	4.442	0.520	2.3%	1.5%
	20	−0.009	3.817	0.564	3.2%	2.6%
	50	−0.059	3.406	0.538	5.2%	4.3%
	100	−0.051	3.108	0.541	7.6%	6.3%
	200	0.042	2.674	0.588	10.9%	8.5%
DeepInception	5	0.121	5.638	0.470	0.7%	0.5%
	10	−0.089	7.107	0.445	1.5%	0.8%
	20	−0.072	6.022	0.523	2.0%	1.5%
	50	0.117	4.651	0.612	2.9%	3.1%
	100	0.231	3.561	0.670	4.3%	4.6%
	200	0.387	2.773	0.673	6.5%	6.5%

3. **Sign agreement is near chance level.** At $k = 20$, sign agreement ranges from 0.523 to 0.576—barely above the 0.5 baseline expected from random polarity assignment. The truncated S_t cannot reliably distinguish between safety-activated and compliance-activated states.
4. **Attack-dependent degradation.** GCG shows the worst preservation: even at $k = 200$, Pearson $r = 0.042$. This is consistent with GCG's token-level optimization distributing probability mass across unusual token positions that rarely appear in the top-k window, making the truncated signal particularly noisy.

Reconciling with the lexicon-internal ablation. The stability reported in the lexicon-internal top-k_{agg} ablation (Appendix L) is *not* contradicted by these results. That ablation showed that *once lexicon tokens are identified in the logit vector*, the score is robust to computing the mean over only 5–10 of them. The vocabulary-level simulation reveals the upstream bottleneck: at typical API truncation levels, too few lexicon tokens are present in the top-k window to compute a meaningful signal in the first place.

Implications. This simulation confirms that S_t—as currently formulated—requires access to the full logit vector (or at minimum, targeted queries for specific token logprobs). Current API offerings ($k \leq 20$) are insufficient. This empirically validates our positioning of S_t as a diagnostic for open-weight model auditing rather than a universal deployment-time detector, and motivates future work on API-compatible reformulations such as targeted logit-bias probing or lightweight distilled classifiers.

U Token Rank Linkage Analysis

To validate that S_t reflects actual token-level rank dynamics, we extract the vocabulary rank of the highest-ranked refusal and compliance tokens at each decoding step alongside S_t, using Llama-3.1-8B-Instruct across two conditions (MCM: $n = 60$; DeepInception: $n = 60$, of which 36 are successful jailbreaks).

Findings. When $S_t < 0$ (refusal-dominant), the best refusal token occupies rank 1–10 in the vocabulary; when $S_t > 0$, refusal tokens drop to rank 100–10,000 (Figure 11). The relationship is monotonic and spans $\sim$4 orders of magnitude, confirming that S_t is not an artifact of aggregation but directly tracks which lexicon dominates the model's next-token distribution. Successful jailbreaks cluster in the $S_t > 0$ region with consistently high refusal ranks (> 100), indicating that refusal tokens are effectively suppressed throughout compliant generation.

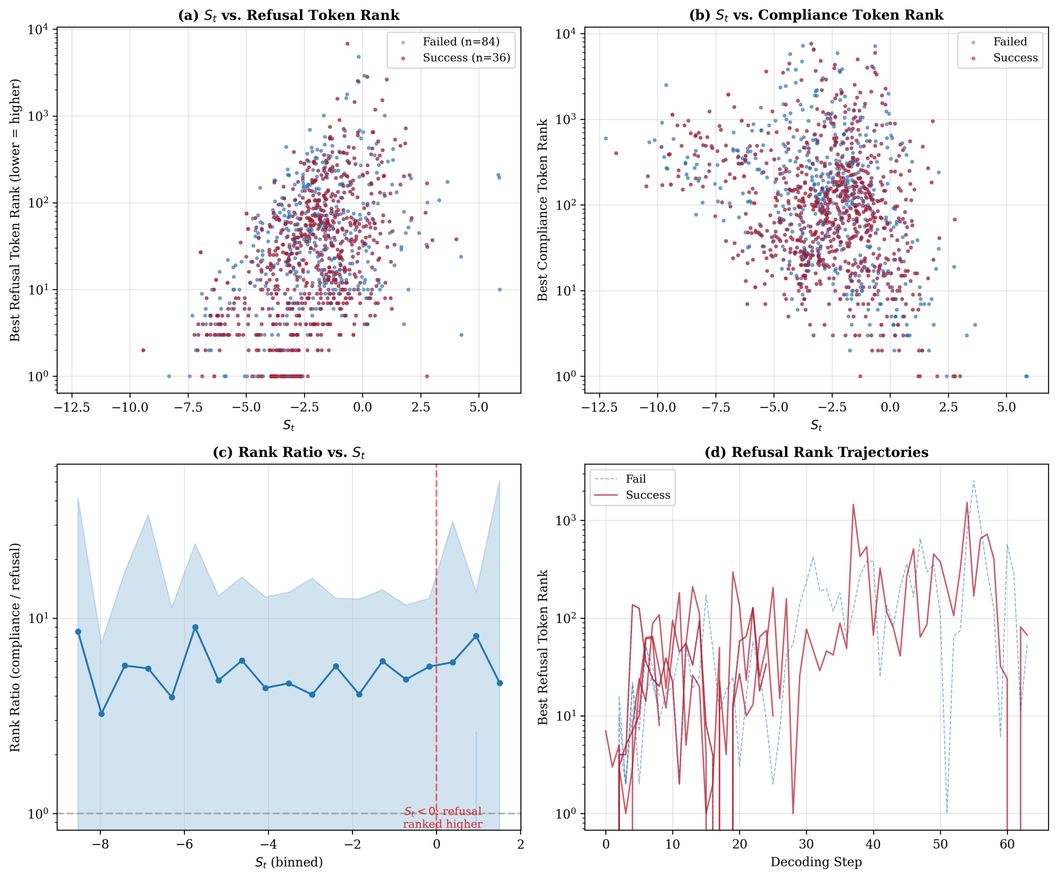

Figure 11: **Token rank linkage.** (a) S_t vs. best refusal token rank: negative S_t corresponds to refusal tokens in top-10; positive S_t pushes them to rank >100. (b) Inverse pattern for compliance tokens. (c) Rank ratio confirms monotonic relationship. (d) Per-sample refusal rank trajectories: successful jailbreaks (red) maintain high refusal ranks throughout.

V Temporal Signal Comparison: S_t, Hidden-State, and Token Rank

We simultaneously extract three temporal signals at each decoding step: (i) S_t (logit-margin score), (ii) hidden-state projection onto the refusal direction at layer 16, and (iii) $-\log_{10}$(best refusal rank) as a token-level indicator. The refusal direction is computed via difference-of-means between benign and harmful hidden states.

Lead/lag analysis. Cross-correlation between S_t and hidden-state projection reveals an asymmetry (Figure 12): for **failed jailbreaks**, peak correlation occurs at lag 0 ($r = 0.70$)—the two signals move simultaneously, consistent with a coordinated refusal response. For **successful jailbreaks**, the peak shifts to lag -1 ($r = 0.60$)—S_t changes *before* hidden-state shifts. This suggests that in successful attacks, the compliance/refusal balance at the logit level reconfigures before deeper representations adjust, providing a mechanistic window for early-step detection.

W Compliance-Dominant Jailbreaks

We systematically examine successful jailbreaks that occur under sustained compliance-dominant token bias ($\bar{S} > 0$).

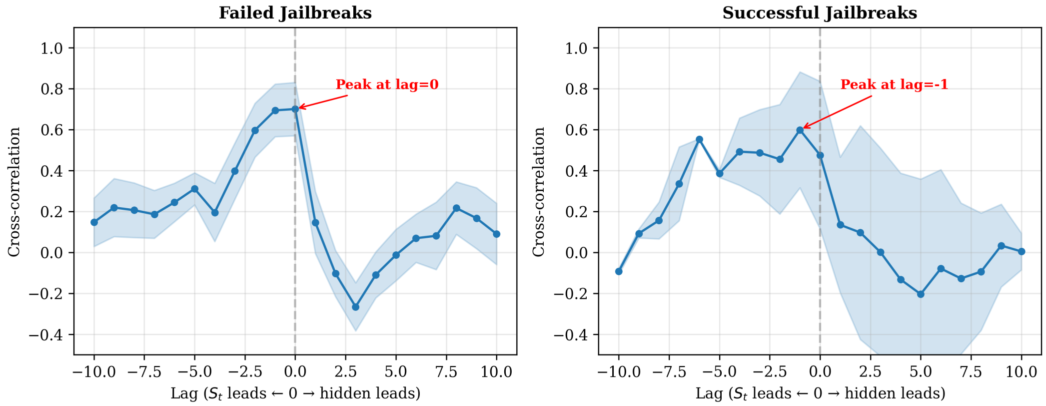

Figure 12: **Lead/lag analysis.** Cross-correlation between S_t and hidden-state refusal projection. Left: failed jailbreaks peak at lag=0 (simultaneous). Right: successful jailbreaks peak at lag=-1 (S_t leads hidden by 1 step).

Prevalence. 91.6% of successful jailbreaks (240/262) maintain $\bar{S} > 0$ throughout generation, meaning compliance lexicon tokens dominate refusal tokens at every step on average. Table 52 reports per-condition breakdown.

Table 52: Percentage of successful jailbreaks with $\bar{S} > 0$ (compliance-dominant generation). Most conditions show 91–100% prevalence; only Qwen+MCM is lower.

Condition	$n_{\mathbf{succ}}$	$n_{\bar{S}>0}$	**%**	$\bar{S}_{\mathbf{mean}}$
Gemma + GCG	20	20	100%	0.98
Gemma + MCM	11	11	100%	0.88
Llama + DeepInc.	33	30	91%	0.59
Llama + GCG	5	5	100%	0.76
Llama + MCM	3	3	100%	0.33
Mistral + DeepInc.	48	48	100%	0.43
Mistral + GCG	34	31	91%	0.28
Mistral + MCM	3	3	100%	0.46
Qwen + DeepInc.	35	35	100%	1.32
Qwen + GCG	46	42	91%	0.71
Qwen + MCM	24	12	50%	-0.01
All	**262**	**240**	**91.6%**	0.61

Interpretation. The high prevalence of compliance-dominant jailbreaks reveals a fundamental limitation of using $\bar{S}$ sign alone: harmful content can be generated using tokens whose logit profile resembles benign compliance. Using $\bar{S}$ sign as a binary detector yields only 8% recall and 9% precision—confirming that S_t's value lies in its *temporal dynamics* (when and how safety engages) rather than in summary-level polarity. This observation strengthens the diagnostic framing: S_t is most informative as a process-level diagnostic that reveals refusal activation timing, not as a standalone classifier. The 91.6% compliance-dominant rate also explains why the Type A/B taxonomy is essential—outcome-only metrics cannot distinguish between "refusal signals emerged but the attack operated through context-level bias" (Type A) and "refusal never engaged" (Undetected).

X Probe-Derived Lexicon Validation

A fundamental question is whether the handcrafted lexicon targets tokens that the model itself associates with safety-relevant behavior, or merely surface-correlated keywords. We address this by projecting all vocabulary tokens onto a learned refusal direction and checking where handcrafted tokens fall in this principled ordering.

X.1 Method

Following Arditi et al. [1], we compute a refusal direction $\hat{\mathbf{r}}$ via difference-of-means between hidden states for 16 harmful and 16 benign prompts at a middle layer. We then project each row of the unembedding (lm_head) matrix $\mathbf{W} \in \mathbb{R}^{|\mathcal{V}| \times d}$ onto $\hat{\mathbf{r}}$:

$$p(v) = \mathbf{W}_v \cdot \hat{\mathbf{r}}, \quad v \in \{1, \ldots, |\mathcal{V}|\}. \tag{12}$$

High $p(v)$ indicates that token v is associated with the refusal direction in the model's output space; low $p(v)$ indicates association with compliance. This provides a *model-intrinsic* ranking of all vocabulary tokens by safety relevance, without any manual selection.

X.2 Results

Table 53 reports the probe-derived rank of single-word handcrafted refusal tokens across all four models. Multi-token phrases (e.g., "I cannot") are excluded because first-token matching maps them to generic tokens ("I") whose projection reflects general frequency rather than safety semantics.

Table 53: Probe-derived rank percentile of single-word handcrafted tokens. **Refusal tokens**: rank percentile from top (100% = rank 1). **Compliance tokens**: rank percentile from bottom (100% = last rank). Higher values indicate stronger alignment with the model's internal refusal/compliance representation.

Token	Llama	Mistral	Qwen	Gemma
Refusal tokens (percentile from top)				
`illegal`	85.7%	**100.0%**	65.9%	99.4%
`harmful`	91.4%	98.1%	97.7%	99.2%
`dangerous`	64.9%	**100.0%**	77.4%	99.2%
`Sorry`	89.0%	89.4%	60.8%	50.3%
`policy`	89.2%	92.0%	87.9%	23.0%
Compliance tokens (percentile from bottom)				
`Sure`	90.0%	37.5%	51.1%	46.9%
`Step`	90.3%	43.6%	95.5%	78.6%
`Below`	50.6%	**98.9%**	93.0%	96.4%
`Procedure`	96.6%	49.3%	70.6%	33.5%
`Instructions`	77.3%	18.4%	37.8%	55.1%

Key findings.

1. **Harm/illegality descriptors are strongly aligned.** Tokens `illegal`, `harmful`, and `dangerous` rank in the top 1–10% of the refusal direction projection across all four models. In Mistral, `illegal` is the single highest-projection token in the entire 32K vocabulary. This confirms that the harm/illegality category of the handcrafted lexicon directly targets tokens that the model's internal safety representation most strongly activates.
2. **Compliance alignment varies but is directionally correct.** Core compliance tokens (`Sure`, `Step`, `Below`) generally fall in the compliance half of the projection ranking, though with more variance than refusal tokens. This is consistent with compliance being a less tightly constrained behavior than refusal—models can comply in many ways, but refuse in stereotyped patterns.
3. **Jaccard overlap is near zero—by design.** The top-25 probe-derived tokens include model-internal artifacts (`<unused>` tokens in Gemma, BPE fragments, multilingual tokens) that would never appear in a handcrafted lexicon. Near-zero Jaccard with high directional agreement demonstrates that the handcrafted lexicon is a *strict subset* of the safety-relevant token space: the probe identifies many additional tokens, but the handcrafted tokens are correctly positioned within the probe ordering.
4. **The probe provides a path to automated lexicon construction.** Selecting the top-N tokens by $|p(v)|$ yields a fully automated, model-specific, language-agnostic lexicon that requires no manual curation. This approach is particularly promising for multilingual models, where handcrafted English lexicons are inapplicable.

Implications for the handcrafted lexicon. The probe validation confirms that the handcrafted lexicon is not arbitrary: its tokens are systematically positioned in the correct direction of the model's internal safety representation. The handcrafted approach sacrifices per-model optimality for interpretability and cross-model generality—a deliberate design choice for a diagnostic tool whose primary contribution is the temporal decomposition framework, not the lexicon itself.